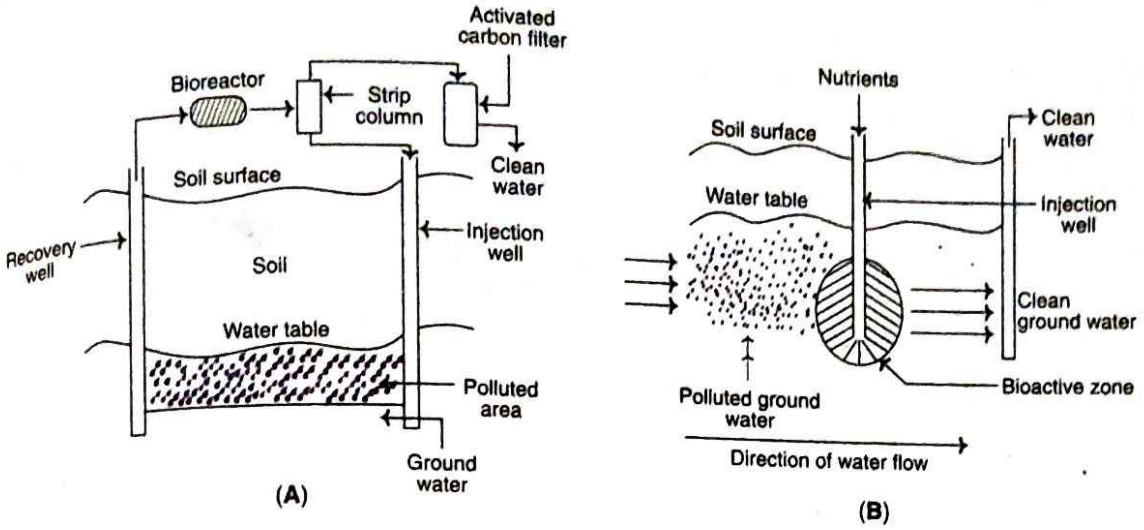


Fig. 59.7 : Bioremediation of ground water (A) Pump- and -treat technique (B) Biofencing technique.

Biofencing technique for bioremediation of ground water

Biofencing is an improved technique for the bioremediation of ground water. It consists of installation of a bioactive zone at the down-gradient edge of a contaminated ground water area. Nutrients are injected through a well to the bioactive zone (***Fig. 59.7B***). As the ground water passes through the bioactive zone (by the impact of natural direction of flow), the pollutants are biodegraded, and clean ground water comes out.



60 Global Environmental Problems

Some of the environmental issues attract world-wide attention, as they are the global problems. These include ***green house effect and global warming***, ***depletion of ozone*** and ***acid rain.*** The global environmental problems along with the environmental sustainability are briefly described in the following pages.

GREEN HOUSE EFFECT AND GLOBAL WARMING

The earth receives solar energy from the sun in the form of short wave radiations (mostly visible light). This solar energy is absorbed by the earth's surface and emitted into the space as long-wave (infra red) radiations. If the incoming radiations from the sun and the outgoing radiations from the earth are equal, then a good balance between the absorbed and emitted energy by the earth is maintained. But this does not happen.

The incoming short-wave radiations (from the sun) can easily pass through the atmosphere while the outgoing long-wave radiations (from the earth) are partially absorbed by certain gases in the atmosphere. ***Green house effect refers to the phenomenon of retention of earth's heat by the atmosphere***. The ***consequence of green house effect is global warming***.

GREEN HOUSE GASES

The ***gases that cause green house effect*** are collectively referred to as green house gases (***Fig. 60.1***). The most important green house gases are ***carbon dioxide*** (55%), ***methane*** (15%), ***chlorofluorocarbons*** (17%), ***nitrous oxide*** (6%) and several other gases like carbon monoxide, nitrous oxide, ozone, sulfur dioxide, fluorine, bromine and iodine (7%). The presence of water vapour in the atmosphere in association with green house gases significantly contributes to global warming.

Sources of green house gases

The important sources of major green house gases are given in the ***Table 60.1***. Fossil-fuel combustion, decompostion of organic wastes, deforestation and industries largely contribute to green house gases. Chlorofluoro-carbons (CFCs) are synthetic chemicals widely used in the preparation of refrigerants, solvents and aerosol propellants. ***CFCs are dangerous environmental pollutants*** that significantly contribute to green house effect. The global warming potential of CFCs is around 7,000 in comparison to CO_2 taken as 1, methane 11 and nitrous oxide 260.

The relative contribution of total green house gases from the major contributing sources is given in the next page.

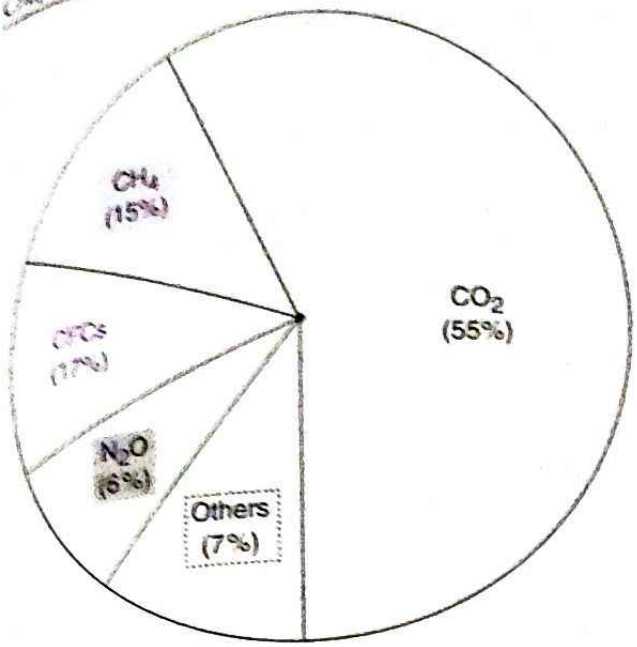


Fig. 60.1 : Approximate contribution of green house gases in global warming (CFCs-Chlorofluorocarbons)

Fossil-fuel combustion and anaerobic decomposition of organics	– 57%
Chlorofluorocarbons	– 17%
Agriculture	– 15%
Deforestation	– 8%
Others	– 3%

Effects of green house gases and global warming

The natural occurrence of green house effect is essential for the existence of life on earth. For instance, it is only due to the presence of CO_2 and water vapour, the mean temperature of the earth's surface is around 17°C. In their total absence, the temperature would have been around –17°C, where life cannot exist!

The main problem is the ever increasing concentrations of the green house gases in the atmosphere that are resulting in irreversible and highly dangerous climatic changes. The effects of global warming contributed by green house gases are listed below.

1. **Increase in temperature :** There occurs a general warming up of the atmosphere due to green house gases. It is estimated that the temperature increases by about 0.3°C per decade.

2. **Changes in sea level :** Due to a rise in temperature, there occurs thermal expansion of oceans and melting of glaciers, ice sheets and ice caps. The net result is an increase in sea level. It is estimated that during the last 100 years, the average sea level has risen by about 20 cm. As per the present prediction, the sea level is expected to rise by 20–30 cm in the next 2-3 decades. Rise in sea levels may be disastrous to some low-lying areas e.g. Netherlands, Maldives.

3. **Water imbalance :** The warmer world would have less water available on the earth and this may lead to water crisis.

4. **Crop yield :** Due to a rise in temperature, the agricultural production may be reduced. There may also occur dislocation of crop lands. Some researchers however, predict that the crop yield may be higher in some areas due to increased availability of CO_2.

5. **Ecosystem :** Some disturbances in the ecosystem and the existence of living organisms may not be ruled out.

Table 60.1 Important sources of major green house gases

Green house gas	*Sources*
Carbon dioxide (CO_2)	Fossil-fuel burning Deforestation Respiration Emissions from industries
Methane (CH_4)	Anaerobic decomposition of organic wastes Gas, oil and coal production Burning of biomass Wetland cultivation
Chlorofluorocarbons (CFCs)	Refrigerants Solvents Aerosol propellants Foam packagings
Nitrous oxide (N_2O)	Fertilizers Natural soils Fossil-fuel combustion
Ozone (O_3)	Photochemical reactions in troposphere
Carbon monoxide (CO)	Fossil-fuel combustion Biomass burning

6. **Human health** : The changing climatic conditions, due to global warming, may adversely affect human health. For instance, the risk of spreading major tropical diseases (malaria, filariasis, schistosomiasis, dengue fever, yellow fever) would be higher.

MEASURES TO CONTROL GREEN HOUSE EFFECT

The major contribution of green house gases comes from the developed countries. Thus, with a world's population of about 15%, these countries contribute to as much as 65% of green house gases. This is mostly because of industrialization and changing life styles of people. It is therefore, natural to expect that the developed countries significantly contribute (finances and resources) for the containment of global warming.

There is a worldwide concern on the alarming impact of global warming due to green house effect. Several steps are being taken at the international and national levels for the protection of environment with particular reference to green house effect.

Some of the important approaches for the management of green house effect with special reference to biotechnology are briefly described.

1. **Renewable forms of energy** : The various forms of energy generation (fossil-fuel burning, fuel wood combustion) are intimately linked to the production of green house gases. Switching over to renewable froms of energy (the best being solar source) is highly desirable.

2. **Reforestation** : Deforestation is major concern. Growing of plants wherever possible and reforestation are the need of the hour. Plants will take up atmospheric CO_2 and generate O_2, and thus significantly help to reduce the green house effect.

3. **Development of energy-efficient industries** : Attention should be directed for the technological advancement of industries with low energy consumption. And wherever possible, renewable sources of energy should be used.

4. **Nuclear power industries** : Cost-effective installation of nuclear power stations is advocated. This will certainly help to solve energy crisis, besides minimizing the green house effect. But care should be taken for a safe disposal of nuclear wastes.

5. **Other energy sources** : Generation of energy from different sources that do not produce CO_2 are desirable. These include hydroelectric energy, solar energy, wind energy, geothermal energy and tidal energy.

6. **Minimal use of CFCs** : As already stated chlorofluorocarbons significantly contribute to the green house effect. They contribute to about 12% of green house gases. CFCs are about 4,000–7,000 times more potent than CO_2 in causing green house effect. For these reasons, many environmentalists advocate a halt for the production of CFCs. This is not practicable, since CFCs have become a part of the life system of a modern man. Hence, their minimal use is advocated.

As is evident from the above discussion, the ***green house effect can be minimized by reducing the addition of CO_2, CFCs, methane and nitrous oxide to the environment.***

THE PROBLEM OF OZONE

Ozone (O_3) is a blue coloured gas with a pungent smell. It forms a thick layer in the stratosphere of the atmosphere (16 to 40 km). The concentration of ozone is in the range of 2-8 ppm (parts per million by volume) depending on the distance.

Generation of ozone

Ozone is produced naturally when oxygen is dissociated by solar ultraviolet radiations (80–240 nm).

$$O_2 \xrightarrow[(80\text{–}240\ nm)]{\text{UV radiation}} O + O$$

$$O + O_2 + M \longrightarrow O_3 + M$$

(M represents a third body molecule like N_2 or O_2).

Beneficial affects of ozone

Ozone is capable of absorbing ultraviolet radiations from the sun (200–300 nm) so that they do not reach the earth and cause health hazards (described later). In this process, ozone dissociation occurs as follows.

$$O_3 \xrightarrow[(200\text{–}300\ nm)]{\text{UV radiation}} O_2 + O$$

Thus ozone is constantly being generated and destroyed in the stratosphere.

DEPLETION OF OZONE

There are a number of pollutant gases in the stratosphere-nitric oxide (NO), nitrous oxide (N_2O), chlorine (Cl) and chlorofluorocarbons (CFCs like chlorofluoromethane or freon) that can react with ozone to produce oxygen. The net result is that the ozone gets depleted in the stratosphere of the atmosphere.

There are three major mechanisms for the destruction of ozone-nitrogen system, hydrogen system and chlorine system.

Nitrogen system

About 60% of the ozone destruction occurs by nitrogen system. Nitrous oxide (N_2O), produced by the microbial action in soils and oceans enters the atmosphere and reaches the stratosphere. Here, in the presence of light, N_2O reacts with nascent oxygen (O) to form nitric oxide (NO). The latter is a powerful destroyer of ozone.

$$N_2O + O \xrightarrow{\text{UV radiation}} 2NO$$

$$N_2O \xrightarrow{\text{UV radiation}} NO + O$$

$$NO + O_3 \longrightarrow NO_2 + O_2$$

Hydrogen system

Approximately 10% of the ozone is destroyed by hydrogen system. The hydroxyl (OH) group that reacts with ozone is mostly derived from water, and to a lesser extent from methane.

The reactions involving H_2O are shown below.

$$H_2O + O \xrightarrow{\text{UV radiation}} 2OH$$

$$OH + O_3 \longrightarrow H_2O + O_2$$

Chlorine system

Chlorofluorocarbons (CFCs) and natural chlorine can also destroy ozone to a significant extent. CFCs on dissociation form Cl which acts on ozone, as shown below.

$$CFCl_2, CF_2Cl_2 \xrightarrow{\text{UV radiation}} Cl_2 + Cl$$

$$Cl + O_3 \longrightarrow ClO + O_2$$

Other systems

Besides the major systems described above, there are several other compounds that can destroy ozone. A group of chemicals namely halons, widely used in fire extinguishers, are important in this regard. The major halons are bromochlorofluorocarbons and bromofluorocarbons.

OZONE HOLE

It was in 1987, the first evidence for the depletion of ozone over the entire Antarctic content came to light. This is regarded as ozone hole, and the depletion of O_3 was observed between 12 to 24 km altitude. Later evidence indicated that the occurrence of ozone hole over Antarctic continent is an annual phenomenon, during August and September. The factors for the causation of ozone hole are not clearly known.

The present belief that the chlorine and chlorine radicals are mainly responsible for ozone hole. It is feared that more ozone holes may develop which is highly dangerous.

EFFECTS OF OZONE DEPLETION

Ozone acts as a filter of ultraviolet radiations from the sun with a result that the hazardous UV rays do not reach the earth. The ***biologically active UV radiations (UV-B)*** are in the range of 2.9×10^{-7} nm to 3.2×10^{-7} nm. They are highly senstive for ozone depletion.

UV-B radiations cause several harmful effects to the living (humans, animals, plants) and non-living (materials) systems on earth. Some of them are briefly described.

Effects on human health

1. The incidence of skin cancer (melanoma) is very high in the population exposed to UV-B radiations. Melanoma is associated with overexposure to sunlight-mostly found in people who spend more time outdoors. It is estimated that with every 1% decrease in ozone layer, there would be an increase of about 3% in the skin cancers of people.

2. UV-B radiations may damage DNA and cause mutations that may result in various types of cancers.

3. Exposure to UV-B radiation is also associated with several other health complications-damage to eyes, decreased immune response, increased incidence of several infections.

Effects on terrestrial plants

1. Exposure to UV-B radiations in some plants may result in reduced growth and smaller leaves, with a reduced efficiency of photosynthesis.

2. The quality and quantity of foods are adversely affected.

3. Retardation in the growth of forests.

Effects on aquatic ecosystems

1. The aquatic life is very vulnerable to UV-B radiations, particularly up to a depth of 20 m in clear waters and a depth of 5m in unclear waters.

2. Harmful effects have been observed in fishes, crabs, shrimp and zooplanktons.

3. There occurs a reduction in the photosynthetic efficiency of phytoplankton.

MEASURES TO CONTROL OZONE DEPLETION

The only effective way of controlling ozone depletion is the complete elimination of the factors responsible for it. As already described oxides of nitrogen, chlorofluorocarbons and halons largely contribute to ozone destruction. These are the environmental pollutants and reduction in their production/utilization will largely help to control ozone depletion.

ACID RAIN

The normal rain water is slightly acidic with a pH in the range of 6-7. This acidity is contributed by the naturally occurring CO_2 which gets dissolved in the rain water to form carbonic acid.

$$CO_2 + H_2O \longrightarrow H_2CO_3$$

The lowest pH of the normal rain is around 5.6.

When the ***pH of the rain water is less than 5.5, it is considered as acid rain***. This acidity is predominantly contributed by two acids namely ***sulfuric acid*** and ***nitric acid***, and to a lesser extent by hydrochloric acid and organic acids.

DEVELOPMENT OF ACID RAIN

As the industrialization worldover increases, there is an increase in the environmental pollution by oxides of sulfur and nitrogen. These gaseous pollutants are light and they can be carried away (from the site of production) to hundreds and thousands of kilometers by the winds. Thus, the export of the acid forming pollutants may occur from one region to another region, and from one country to another country. For this reason, acid rain is a global problem.

The following reactions summarise the formation of acid rain from the oxides of sulfur (SO_x) and oxides of nitrogen (NO_x).

$$SO_x + O_2 + H_2O \longrightarrow \underset{\text{(Sulfurous acid)}}{H_2SO_3} + \underset{\text{(Sulfuric acid)}}{H_2SO_4}$$

$$NO_x + O_2 + H_2O \longrightarrow \underset{\text{(Nitrous acid)}}{HNO_2} + \underset{\text{(Nitric acid)}}{HNO_3}$$

In the development of acid rain, there occurs diffusion of SO_2, and diffusion of HNO_3 gas into the cloud particles/droplets.

EFFECTS OF ACID RAIN

Acid rain disturbs the environment. The actual effects of acid rain depends on the degree of the acidity and the nature of the environment-aquatic, terrestrial, human population and materials. Some of the important aspects are described.

Effects on aquatic environment

The aquatic system is very sensitive to acid rain. With periodical falls of acid rains, the lakes become more and more acidic.

1. The fishes and other aquatic organisms including microorganisms are very sensitive to pH. At a pH below 4, most of them die.

2. The surving fishes contain high concentrations of toxic metals such as mercury, aluminium, lead and zinc (these metals find their entry into lakes by leaching of surrounding rocks by acid rain).

Effects on terrestrial environment

1. Acid rain damages forests and all forms of vegetation.

2. Soil acidification, due to acid rain, may lead to necrosis of leaves.

3. Plants also accumulate several toxic metals e.g. aluminium, cadmium, mercury, lead.

Effects on human health

1. Sulfuric acid of acid rain is very dangerous, as it causes breathing problems.

2. The toxic metals (Cd, Zn, Hg, Cu, Al) liberated from rocks and soils by acid rain may ultimately reach the human body through plants and animals through food chain or drinking water. The toxic effects of these metals on humans are well known.

Effects on materials

The acid rain causes deterioration of buildings, marble rocks, limestones etc. Thus, acid rains have become a big threat to preservation of monuments and cultural heritages throughout the world.

MEASURES TO CONTROL ACID RAIN

The most effective measure to control acid rain is to drastically reduce the production of acid-forming gases i.e. the ***industrial emission of oxides of sulfur and nitrogen should be minimized.***

Some measures have been taken to reduce the SO_2 emission into atmosphere-desulfurification of the fuels used in industries, recovery of SO_2 as H_2SO_4.

More concrete measures need to be taken globally for effective control of acid rain.

ENVIRONMENTAL SUSTAINABILITY AND BIOTECHNOLOGY

The various aspects of environment and their impact on earth with special reference to human beings have been described (Chapter 54). Some of the important global environmental problem are described in this chapter.

Despite the occurrence of a complex and sometimes dangerous alterations in the environment, the life processes should go on, normally as far as possible. Environmental sustainability broadly deals with ***the sustenance (support and maintenance) of the environment to continue life on earth in a normal way.***

Biotechnology significantly contributes to environmental sustainability. There are three different approaches in this regard.

1. **Pollution protection :** Environment can be protected from pollution by using renewable raw materials, recycling the processes, appropriate methods for degradation of wastes.

2. **Pollution clean-up :** This can be done by employing biotechnological methods for clean up of oil spills, detoxification of contaminated soils, and purification of water supplies.

3. **Pollution control :** Once pollution has occurred, biotechnological methods can be used for control. For instance, heavy metals can be recovered from polluted water.

It is a fact that the various activities of humans are exceeding the sustainable capacity of the earth. Increased industrialization and urbanisation are adversely affecting the environment (with heavy pollution).

The biotechnologists are expected to improve the existing biological, chemical and various other industrial processes to make them environmental friendly. They should develop integrated systems which are efficient, controllable and clean. It is the duty of the environmental biotechnologists to achieve a maximum global environmental safety.

Eco-Tech : This is a new environmental technology concept. International Organisation for Biotechnology and Bioengineering in 1994, defined Eco-Tech as follows :

"Embedding technology into ecosphere and human culture by using the whole range of biodiversity in a holistic and low-invasive way in order to achieve benefits for humankind obeying ecological principles".

SECTION IX

BIOTECHNOLOGY AND SOCIETY

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61 Biotechnology – Society, Risks, Ethics, Patenting

Advances in biotechnology, and their applications are most frequently associated with controversies. Based on their perception to biotechnology, the people may be grouped into three broad categories.

1. ***Strong opponents*** who oppose the new technology, as it will give rise to problems, issues and concerns humans have never faced before. They consider biotechnology as an unnatural manipulative technology.

2. ***Strong proponents*** who consider that the biotechnology will provide untold benefits to society. They argue that for centuries the society has safely used the products and processes of biotechnology.

3. A ***neutral group*** of people who have a balanced approach to biotechnology. This group believes that research on biotechnology (with regulatory systems), and extending its fruits to the society should be pursued with a cautious approach. The risks and benefits of the developments of biotechnology may not be much different from that of any other branch of science.

BENEFITS OF BIOTECHNOLOGY

The fruits of biotechnology are beneficial to the fields of healthcare, agriculture, food production, manufacture of industrial enzymes and appropriate environmental management.

It is a fact that modern technology in various forms is woven tightly into the fabric of our lives. Our day-to-day life is inseparable from technology. Imagine life about 1-2 centuries ago where there was no electricity, no running water, sewage in the streets, unpredictable food supply and an expected life span of less than 40 years. Undoubtedly, technology has largely contributed to the present day world we live in. Many pepople consider biotechnology as a technology that will improve the quality of life in every country, besides maintaining living standards at a reasonably higher level.

The probable positive and negative effects of biotechnology, with special reference to developing countries are given in ***Table 61.1***.

ELSI OF BIOTECHNOLOGY

Why so much uproar and negativity to biotechnology? This is mainly because the major part of the modern biotechnology deals with genetic manipulations. These unnatural genetic manipulations, as many people fear, may lead to unknown consequences.

ELSI is the short form to represent the ***ethical, legal and social implications of biotechnology***. ELSI broadly covers the relationship between biotechnology and society with particular reference to ethical and legal aspects.

Table 61.2 The probable positive and negative effects of biotechnology (particularly in the developing countries)

Positive effects

- Improved health and life span (through more effective pharmaceuticals, vaccines etc.).
- Increased crop production and more food.
- Decreased dependence on import of food, fertilizers and chemicals.
- Enhanced biomass production for energy.
- Use of plants and animals as bioreactors.
- Improved food storage facilities.
- Increased production and health of livestock.

Negative effects

- More dependence on developed countries and private companies for technology and resources.
- Increase in unemployment as the labour requirement is less.
- Reduction in natural biodiversity and natural ecosystems.
- More legal and financial complications.
- Increased growing of cash crops at the cost of food crops.
- Production of more and more herbicide resistant plants.

Risks and ethics of biotechnology

The modern biotechnology deals with genetic manipulations of viruses, bacteria, plants, animals, fish and birds. Introduction of foreign genes into various organisms raises concerns about the safety, ethics and unforeseen consequences. Some of the popular phrases used in the media while referring to experiments on recombinant DNA technology are listed.

- Manipulation of life
- Playing with God
- Man-made evolution

The major apprehension of genetic engineering is that through recombinant DNA experiments unique microorganisms or viruses (either inadvertently or sometimes deliberately for the purpose of war) may be developed that would cause epidemics and environmental catastrophes. Due to these fears, the regulatory guidelines for research dealing with DNA manipulation were very stringent in the earlier years.

So far, risk assessment studies have failed to demonstrate any hazardous properties acquired by host cells/organisms due to transfer of DNA. Thus, ***the fears of genetic manipulations may be unfounded*** to a large extent. Consequently, there has been some relaxation in the regulatory guidelines for recombinant DNA research (For more details, Refer Chapter 6).

It is now widely accepted that biotechnology is certainly beneficial to humans. But it should not cause problems of safety to people and environment, and create unacceptable social, moral and ethical issues. The public fears of biotechnology, besides some of the risks, social and ethical issues can be better understood with special reference to the following.

1. Therapeutic products for use in healthcare.
2. Genetic modifications of foods and food ingradients and their consumption.
3. Release of genetically engineered organisms into the environment.
4. Applications of human genetic research.

RECOMBINANT THERAPEUTIC PRODUCTS FOR HUMAN HEALTHCARE

It is fortunate that there is ***no serious criticism*** about the use recombinant products for medical applications. This is mostly because the therapeutic products and strategies are designed to cure diseases, alleviate sufferings and improve the quality of life. Further, the products are used under the medical supervision. In general, the recombinant products designed for human healthcare are more readily acceptable by the public. Good examples are the use of insulin, interferons, tissue plasminogen activator and various vaccines (for more details, Refer Chapters 15 and 16).

GENETIC MODIFICATIONS AND FOOD CONSUMPTION

The overall objectives of genetic modifications with reference to foods are listed

- To increase the quality and quantity of existing foods.

- To produce new products.
- To improve the financial returns.

Each country has its own regulations for introducing foods into the market. For instance, in USA, the Food and Drug Administration (FDA) regulates the introduction of foods, drugs, pharmaceuticals, and medical devices into the marketplace.

As regards the foods and food ingradients developed by genetic engineering, the FDA believes that no new regulations are needed. The existing regulations for the assessment foods for safety by toxicity, allergenicity and impurity testing are adequate. If the chemical composition of the existing food is altered by genetic modification, it should be specified, and the new product should be accordingly labeled.

To highlight the public perception of genetically modified foods, some selected examples of food ingradients and their acceptance, or certain controversies related to their use, are briefly described.

Chymosin

Chymosin is milk clotting proteolytic enzyme that hydrolyses the milk protein casein to produce curd, which in turn is processed into **cheese**. Traditionally, chymosin is derived from the stomach of calves in the form of rennet.

By genetic engineering techniques, the chymosin gene was cloned and expressed in *E. coli*. This resulted in a large scale and cost-effective production of chymosin. The chemical composition, structure and biological activity of recombinant chymosin were identical to the chymosin of rennet. FDA gave license to chymosin for its commercial use, and it is now widely used in cheese making.

Tryptophan

In 1989–90, an unusually high incidence of the disease **eosinophilia-myalgia syndrome (EMS)** was reported in USA. EMS is a rare disease with muscular pain and respiratory complications, and may be fatal. Investigations revealed that the victims of EMS were consuming large quantities of food supplement tryptophan (obtained from one company). This tryptophan was produced by genetically engineered microorganisms. Chemical analysis of the commercial preparations revealed the presence of certain metabolic derivative of tryptophan. The most important among them was 1-1′ ethylene — bis (tryptophan) which was responsible for EMS (based on the results of animal experiments). It was concluded that the pharmaceutical company did not take adequate care for purification of tryptophan. Consequently, recombinant tryptophan (even without the impurities!) was banned for human consumption in USA.

Bovine somatotropin story

Bovine somatotropin (BST, also known as bovine growth hormone), when injected to dairy cattle, increases the milk production significantly. By recombinant DNA technology, the gene for BST was cloned and expressed in *E. coli*. The recombinant BST (rBST) so produced, when injected to cows was found to increase milk production by 20–25%.

The effect of rBST was considered from two aspects — on the animals and consumers.

1. **Effect of rBST on animals :** Administration of rBST can produce localized swelling at the site of injection. There may be some other adverse effects like increased susceptibility to infection, decreased reproductive capability. The proponents of rBST argue that these problem could occur even in normal animals.

2. **Effect on human health :** The natural BST or even rBST increases the body levels of ***insulin-like growth factor-I*** (IGF-I) which in turn enhances milk production. There is evidence that IGF-I stimulates growth of cancer cells. This causes concern among the consumers of milk produced by using rBST. The counter argument is that rBST injection, after about 100 days of lactation begins, is not associated with increased levels of IGF-I. The opponents of rBST strongly feel that since milk is consumed by most people, any inherent risk, however small is unacceptable.

Recombinant BST was licensed in USA, for use in dairly cattle in 1994. Some countries in fact have banned the use of rBST. There are some people (particularly among the scientists) who believe that the hue and cry raised against rBST is more due to economic and political reasons. It is feared that by rBST use, the dairy industry may be controlled by large industrial groups and the small dairy farms may become unprofitable.

The rBST story is an interesting illustration of the problems surrounding the use of genetic engineering.

RECOMBINANT FOODS AND RELIGIOUS BELIEFS

Some of the ethical concerns of the use of recombinant foods are related to religious beliefs, besides food habits. For instance

- Transfer of pig genes into sheep may offend the sentiments of Jews and Muslims.
- Introduction of animal genes into food plants may invite opposition by strict vegetarians.
- Transfer of human genes to food animals may be unacceptable to some people.
- Feeding of human gene — containing organisms to animals sounds in bad taste (at present, the genetically modified yeast that produce recombinant proteins after their use, are fed to animals).

The religious groups, in general, are selective about the foods to be eaten. However, they are not so rigid when it comes to the use of medically-derived products. For instance — Jews and Muslims may accept pig-derived insulin for use in diabetic patients. This is due to the fact that ***all religious faiths consider human life is the most valuable, and its preservation the first priority***. Further, the general belief is that the human body is violated only by oral consumption and not by injection or surgical interventions.

Eating genes everyday!

Everyday, we eat plants and animals and various products derived from them, besides a large number of bacteria. In other words, we regularly consume genetic material, the DNA, organized into genes in various organisms! So far no one has attempted to categorize genes as vegetarian and non-vegetarian, as we do for foods!!

ARE GM FOODS SAFE?

The production of transgenic plants and animals by genetic engineering techniques has now become routine. These organisms will enter the food chain in the form of ***genetically modified foods*** (GM foods). Some social and environmental groups are against the consumption of GM foods. These people insist that the GM foods should be specifically labeled.

As regards the safety of GM foods, opinions range from one extreme that they are absolutely safe, and improve human nutrition to the other that they should not be consumed at all. Most of the people have opinions somewhere between these two extremes.

RELEASE OF GENETICALLY ENGINEERED ORGANISMS

The release of genetically engineered organisms (***GEOs***), also called as ***genetically manipulated organisms (GMOs)*** into the environment has been a controversial issue; and continues to be so. It is feared that the release of GEOs into the environment could have far-reaching consequences. This is due to the fact that the living GEOs proliferate, persist, disperse, and sometimes may transfer their DNA into other organisms. It is further feared that there exists a possibility of GEOs displacing the existing organisms, besides creating new species. This may lead to severe environmental damage.

For the reasons stated above, the regulatory authories are very careful in permiting the field trails of GEOs. Further, the release of GEOs into the environment has to be carefully monitored and recorded.

Ice-minus *Pseudomonas syringae*

A genetically modified strain of ***Pseudomonas syringae*** was the first GEO that was given permission for field trails (in 1987). This organism is a genetically engineered ice-minus strain, when sprayed onto the leaves could prevent frost damage to the plants (Refer Chapter 50).

Field trails with other GEOs

A number of open-field trails have been conducted with several GEOs during the past two decades or so. The studies concluded that the genetically modified micororganisms do not persist in environment for long, do not transfer the genes into other organism and do not exhibit any abnormal biological functions. Thus, the ***initial apprehensions on the use of GEOs appear to be unfounded.***

Release of transgenic plants and animals

The transgenic plants developed for higher quality and quantity of foods, in general, are more liberally permitted to go to fields. It is generally believed that the transgenic plants do not significantly differ from the natural cultivars (traditional plants) obtained by plant breeding experiments.

The ***transgenic Bt-plants*** such as cotton, corn, soybean and potato were ***approved for cultivation in USA***. However, some countries did not allow Bt-plants in their fields e.g. Bt-rice was not allowed in Philippines, Bt-cotton in France.

The transgenic animals have not posed as big a problem as the transgenic plants. This is due to the fact that animals can be much easily identified and contained (since copulation is animals can be done as derised, unlike in plants where pollination is difficult to control). A majority of transgenic animals are used for medical purposes, hence they are generally appreciated. But the major problem for transgenic animals comes from animal activists.

Many people, and also some Governments, are suspicious of the use of GEOs due to various reasons-risks, societal beliefs and economic concerns.

Biological warfare

The most serious hazard associated with genetic engineering is the construction of harmful biological agents (viruses, microorganisms) either deliberately or otherwise. However, so far there have been no records of any new infectious agents created by recombinant DNA technology.

Most of the countries of the world are signatories to the Biological Weapons Conventions of 1972. As a signatory, a nation pledges 'never to produce microbial or other biological agents, or toxins, whatever may be their method of production, for use in wars'. Many people are, however concerned about the possible use of gene manipulations for military purposes.

APPLICATIONS OF HUMAN GENETIC rDNA RESEARCH

The ultimate goal of advances in biotechnology is for the benefit of mankind (either direct or indirect). Biotechnology largely contributes to human genetic research involving the following areas

- Genetic testing and screening for diseases
- Genetic portifolios
- Human gene therapy.

Genetic testing and screening

Techniques are now available for prenatal testing to specifically detect whether a fetus carries genetic defects. This will help the parents to be better prepared for the future baby.

The negative aspect of prenatal testing is that the couple may opt for abortion even for a minor genetic defect or sometimes for gender bias.

Genetic portifolios

The elucidation of the entire human genome sequence and identification of genes has now become a reality. It may soon be possible to have individual genetic portifolios that will diagnose future health complications e.g. risk for cancer, heart disease. The ***genetic portifolios*** (based on the genes) ***will fortell the individuals' future*** which is now being predicted through stars (astrology).

Genetic portifolios of individual may pose certain problems with regard to marriages, insurances. Who would like to be a spouse of someone who will soon be a victim of cancer or heart attack? Which insurance company would insure a person with a very high risk of diseases? Many ethical committes are of the opinion that insurance companies should not require, or should not be allowed to have access to individuals' genetic portifolios.

Human gene therapy

Theoretically, correction of genetic defects is possible by gene therapy. The present status of human gene therapy has been described (Chapter 13). From the ethical perspective, gene therapy involving introduction of genes into a patient is comparable to the practice of transplation of organs (e.g., heart, liver, lungs). Therefore, there is not much controversy over gene therapy, as long as it is intended to be used to alleviate serious medical disorders. However, the gene therapy must be under a close supervision to satisfy medical, legal, ethical and safety implications, besides addressing the public concerns.

Germ line gene therapy : This is not being carried out at present due to technical, ethical and social reasons. Manipulation of germ cells will lead to serious problems and complications.

At one international meeting on biotechnology the following was the final message given to biotechnology companies on the applications of genetic engineering to humans.

'***Provide the information and listen to the public***'.

HUMAN EMBRYONIC STEM CELL RESEARCH

The human embryonic stem cell (ESC) lines were established in November 1998. These cells are capable of giving rise to any human cell type. ESC lines open the possibilities of treating diseases with cell therapy. Disorders that involve the loss of normal cells such as diabetes mellitus, Parkinson's disease, Alzheimer's disease could perhaps be corrected with cell therapy. This may however, take more time.

There are many ethical and legal issues involved in ESC research, besides several objections raised by the public. In fact, the U.S. Federal Government has banned the use of federal funds for human embryo research for over 20 years. At present, most of research on ESC is being supported by private companies.

CLONING HUMANS ?

After the cloning of the sheep Dolly (in February, 1997), some groups of researchers naturally became interested to eplore the possibilities of cloning humans. The very thought of human cloning has become a highly charged and controversial issue.

A such, considering the biomedical ethics, most of the countries have banned research related to human cloning. It cannot be predicted at present whether some day human cloning may become invitable!

Scientists who were awarded Britain's first licence for human cloning have recently (2005) created an early stage of human cloned embryo by using nuclear transfer. The researchers, as such, are not interested to make babies by cloning. Instead, they wish to create test-tube embryos to supply stem cells that can give rise to every tissue in the body. By this approach, it might be one day possible to repair tissue damages and cure many diseases e.g. Parkinson's disease, diabetes mellitus.

PATENTING BIOTECHNOLOGY INVENTIONS

Biotechnology involves the production of a large number of commercial products of economic importance. Broadly, biotechnology inventions can be categorized into two forms — products and processes.

BIOTECHNOLOGY PRODUCTS

1. **Living entities** : The various forms of life-existing and life-supporting systems derived from living organisms are regarded as living entities. These include microorganism, animals and plants, cell lines, organelles, plasmids and genes.

2. **Naturally occurring products** : The primary and secondary metabolites produced by living systems e.g. alcohol, antibiotics.

BIOTECHNOLOGY PROCESSES

Biotechnology involves development of several processes — isolation, purification, cultivation, bioconversion etc. Novel, innovative, simple and cost-effective processes are developed for the isolation and creation of the above products. Some examples are listed

- Production of monoclonal antibodies
- Bioconversion of sugar to alcohol
- Microbial production of antibiotics.

WHAT IS A PATENT?

A patent is a government (or patent office) — issued document that allows the holder (patentee) the exclusive right to manufacture, use, or sell an invention for a defined period, usually 20 years. The patent may be regarded as a ***legal document safeguarding the previleges and rights of an invention/inventor***.

The very purpose of patenting in biotechnology is to ensure the just financial returns for those who have invested heavily — the finances, intellectual abilities, besides the hard work.

INTELLECTUAL PROPERTY RIGHTS

Patent is the most important form of ***intellectual property rights*** (***IPRs***) in biotechnology. The other IPRs are trade secrets, copyrights, and trade marks.

Trade secrets : The private information pertaining to specific formulations and technical procedures that a company wishes to protect from others are regarded as trade secrets. The best known trade secret is the formula for coca-cola. Only five persons in the world are said to know the formula which has been kept in a bank in Atlanta, Georgia. The trade secret of coca-cola has remained a secret for more than 100 years!

Copyrights : The protection of authorships of published work comes under copyrights of IPRs.

Trade marks : The specific symbols or words to identify a particular product or process of a company constitute trade marks.

THE PROCESS OF PATENTING

As regards patenting of biotechnology inventions, the ***products or processes*** (described on p. 744) are ***patentable***. For patenting, the following criteria must be satisfied.

- The invention must be novel
- It must be useful
- The patent application must provide the full description of the invention.

In contrast to various other technology inventions, biotechnology inventions most often relate to living materials. For this reason, the patenting of biotechnology inventions is often associated with difficulties. Patenting, in general, is a commercial decision, assisted by legal advice.

The patent application must be prepared by an expert with the assistance of a patent lawyer. It must contain the detailed information about the invention — background, description explaining the nature of the invention with figures and illustrations, its utility etc. The patent office, after a thorough scrutiny of the application, may reject or grant the patent. If rejected, the applicant may appeal to the Patent Appeals Board. If again rejected, the decision may be challenged legally. Many a times, the process of patenting becomes a frustrating experience. Thomas Edison, probably the only individual in the world with more than 1000 patents to his credit, once said about patenting 'an invitation to a lawsuit'.

Patenting in different countries

Patenting laws are very complex and vary between different countries. It is interesting to note that a patent application rejected in one country may be given patent in other country. The best example is the production of recombinant human tissue plasminogen activator (tPA) in *E. coli*. Genentech company was awarded patent for tPA production in USA, while it was first rejected in UK. After appeals and legal battles, patent was finally granted in UK also.

Patenting microorganisms

A genetically engineered strain of the bacterium *Pseudomonas* called ***superbug*** was ***the first micro-organism to be patented***. Superbug is capable of breaking down the multiple impuries of crude oil. It took almost 10 years of procedural delays, and legal battles to get the superbug patented. Now, many microrganisms are patented.

Patenting multicellular organisms

A genetically manipulated mouse namely the ***oncomouse*** was the first and probably, to date the only ***animal to be patented*** in both USA and UK. This transgenic mouse carries a gene that makes it susceptible to tumor formation.

Patenting genes?

The human genome projects (HGPs) have elucidated almost the entire sequence of human genome, besides identifying the genes. Patenting of DNA sequences and genes has become a controversial and debatable issue. Some researchers and institutions in fact have approached the patent offices for grant of patents. So far, patents have not been granted for genes. The most important reason being that the genes are natural, universal and inherent biological functional units of all individuals.

Patenting, and cell/tissue donor's rights

While dealing with human tissues/cells, the involvement of the donor of source material becomes very important. John Moore was a victim

of hairy-cell leukemia, a rare form of cancer. His spleen was removed, and Mo cell lines developed. These cell lines were patented by the researchers and their associated organizations. Moore, however was not involved in the patenting.

Moore filed a suit in the court on the ground that he should also be a party to the profits derived by using his tissue. The court allowed his claim to share the profits.

PLANT BREEDERS' RIGHTS

Agriculture for the first time was included (in 1994) in the ***trade-related intellectual property rights (TRIPS)***. TRIPS is a major concern for developing countries.

The plant varieties in many countries (not in India) are protected through plant breeders' rights (PBR) or ***plant variety rights (PVR)***. PVR provides legal protection to the original breeder or owner of the plant variety. It is believed that PBR will ***encourage innovative research and plant breeding programmes***, in view of the expected financial returns.

Plant breeders' rights are comparable to the patent rights. Under TRIPS agreement, plant breeder possesses the exclusive rights over the plant developed. It prevents the third parties from using the plant without the owner's consent.

Although PBR will increase competition and development of good plant varieties, the owner's motto would be profit making. Therefore, PBR will prohibit the free exchange of plant materials, besides threatening the farmer's rights. It is feared that PBR will benefit rich farmers and developed countries only.

PATENTING AND BIOTECHNOLOGY RESEARCH

In the earlier years, research used to be mostly for academic interest and worthwhile scientific contributions. And researchers were purely academic-oriented with not much interest in financial gains from what they do. Further, unlike now, the research used to be supported mostly by Governments/Universities.

There used to be no secrecy in research. Academic recognition and outstanding research publications (open to all) were adequate to satisfy the researchers. Watson and Crick who discovered DNA structure never had a thought of patenting!

The situation has now changed. A good proportion of research is either being conducted or financially supported by private companies/ universities. It is quite natural for these companies to expect returns for their investments. There is a lot of change in the attitude of researchers also. For many scientists, ***financial gains have become more important than academic recognition***. Consequently, some people carry out research secretly and opt for patenting of their discoveries rather than publishing.

We have to accept the fact that the progress of research is usually much faster in private companies compared to many government organizations. This may be due to the higher financial returns and efficient administration.

Probably, the biotechnological research might not have progressed to the same extent as it is today without the dreams of getting the patents by scientists and research funding organizations.

BIOTECHNOLOGY AND THE DEVELOPING COUNTRIES

A major proportion of research related to biotechnology is carried out by the developed countries. The fruits of biotechnology are probably, more useful and relevant to developing countries, as illustrated with the following applications

- Increased levels of nutrition with improved nutrient composition in the foods (through transgenic plants).
- Prevention of child deaths by appropriate immunization (using recombinant vaccines).
- Supply of clean drinking water, and improved sewage disposal (by appropriate bio-technological treatments).

Despite the known benefits, some of the developing countries are reluctant to open doors for the advances made in biotechnology. This may be more due to political considerations rather the economic reasons.

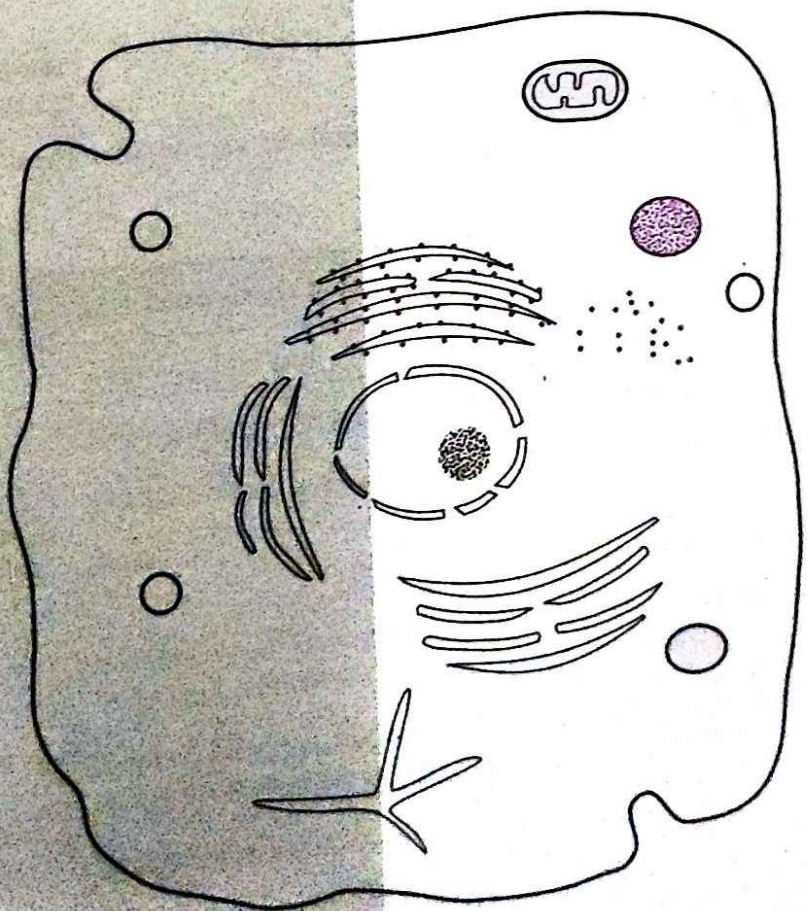
The applications of biotechnology, particularly to human healthcare, may someday be regarded as a barometer to evaluate the progress of a nation.

SECTION X

BASICS TO LEARN BIOTECHNOLOGY

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63 Microorganisms

Microbiology (Greek, mikros—small; bios–life) is the ***science of small*** or ***microscopic organisms***. The most important microorganisms relevant to biotechnology include bacteria, fungi, and viruses. Microorganisms are very widely distributed, and are found almost everywhere in nature. In general, the conditions for their growth and multiplication (food, temperature, moisture etc.) are similar to that of humans. Hence, they are most abundantly present at places where people live.

A very brief description of the microorganisms, relevent to biotechnology is given in this chapter.

BACTERIA

Bacteriology, the study of bacteria, forms a major part of microbiology. The population of bacteria exceeds all other organisms. For instance, one kg of a soil may contain more bacteria than the entire human population! Bacteria are very important in biotechnology for the following reasons

- Diseases they cause
- Domestic uses
- Industrial applications
- Agricultural processes.

CHARACTERISTICS OF BACTERIA

Bacteria are ***prokaryotic unicellular organisms***. They lack organized nucleus, but possess a rigid cell wall comparable to that found in plants. The average size of a bacterium is around 2 µm. The bacteria may be spherical, rod-like, spirally coiled or filament like. Certain bacteria may occur in more than one form. A typical structure of a rod-shaped bacterium is depicted in ***Fig. 63.1***.

Gram-positive and Gram-negative bacteria

Based on the response to Gram's stain, the bacteria are grouped as Gram +ve or Gram –ve. By Gram's stain, several distinguishing features of bacteria can be identified. For instance, Gram +ve bacteria possess single-layered cell wall while Gram –ve bacteria have a double-layered one.

Aerobic, anaerobic and facultative bacteria

On the basis of respiration (i.e. response to O_2), the bacteria are grouped into three categories

1. **Aerobic bacteria :** These bacteria require O_2 for their growth e.g. *Pseudomonas* sp, *Mycobacterium* sp.

2. **Anaerobic bacteria :** These bacteria do not require O_2 to obtain energy, and to grow. In fact, the presence of O_2 is toxic to them e.g. *Peptococcus* sp.

3. **Facultative bacteria :** The bacteria that can grow in both aerobic and anaerobic conditions are regarded as facultative bacteria e.g. *Shigella* sp, *Salmonella* sp.

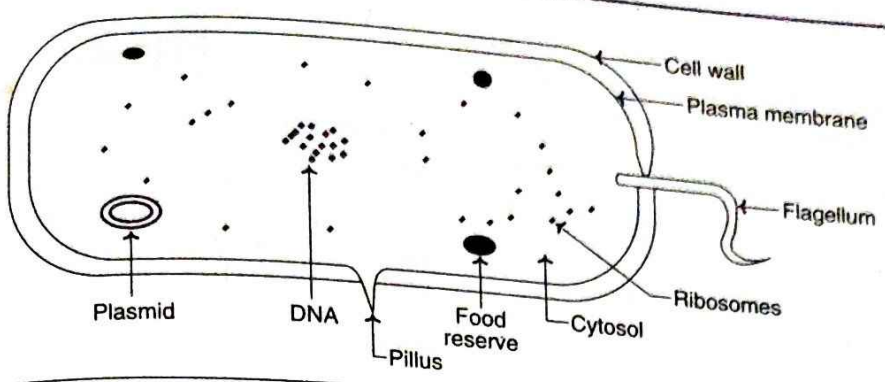


Fig. 63.1 : A typical structure of a rod-shaped bacterium.

Nutritional aspects of bacteria

Based on their nutrition, the bacteria are categorized as autotrophic or heterotrophic.

1. **Autotrophic bacteria :** These bacteria are capable of synthesizing their own food from inorganic substances. They are compable to higher plants in this aspect. Autotrophic bacteria utilize different hydrogen compounds (not H_2O as in the case of higher plants). These include hydrogen, ammonia, hydrogen sulfide and methane e.g. *Hydrogenomonas* sp, *Nitrosomonas* sp, *Methanomonas* sp.

2. **Heterotrophic bacteria :** These bacteria cannot synthesize their own food, and are therefore dependent on the outside source. Heterotrophs are of two types—sporophytes and parasites.

Sporophytes obtain their food from sources of animal or plant origin. These include organic remains like corpses, animal excreta, meats, fruits and various other products of plant and animal origin. Sporophytes secrete digestive enzymes that break the complex organic molecules into simpler and easily absorbable forms. These heterotrophic bacteria are ***useful for the disposal of sewage***, cleansing of leather, and manufacture of certain compounds (alcohols, organic acids). Sporophytes can also spoil foods, and damage soils (by denitrification).

Parasites are the bacteria that obtain their food from living organisms, namely the hosts. They may be either harmless (non-pathogenic) or harmful (pathogenic) to the hosts. *E. coli* is a good example of non-pathogenic bacteria which has a symbiotic relationship in the human intestine. The pathogenic bacteria may cause serious diseases either by destroying the host's cells or releasing toxins e.g. *Clostridium tetani*.

IMPORTANCE OF BACTERIA

Bacteria play a very important and crucial role in the continuous sustenance of life on earth. They are both friends and enemies of humans. Bacteria are important due to the diseases they cause, industrial and agricultural applications.

Diseases caused by bacteria

The bacteria pose a serious threat to humans, animals and plants due to various diseases they cause. A selected list of the important diseases and the corresponding organisms is given in ***Table 63.1***.

Industrial uses of bacteria

An important aspect of biotechnology is the industrial use of bacteria for the production of several compounds. Traditionally, bacteria are in use for the preparation of curd, vinegar, pickles, curing tea and tobacco leaves. In recent years, large-scale production of amino acids, organic solvents, vitamins and antibiotics is being carried out by employing bacteria.

Bacteria play a significant role in the decay and disposal of dead plants and animals. Appropriate sewage disposal requires bacteria. Bacteria are also needed for the disposal of animal dung and human excreta.

Agricultural applications of bacteria

Nitrogen fixation by converting free nitrogen into nitrogenous compounds is very important in agriculture. This is exclusively carried out by

TABLE 63.1 A selected list of diseases caused by microorganisms

Diseases of humans	*Pathogenic microorganism*
Tuberculosis	*Mycobacterium tuberculosis*
Leprosy	*M. leparae*
Typhoid	*Salmonella typhi*
Cholera	*Vibrio cholerae*
Diptheria	*Corynebacterium diphtheriae*
Syphillis	*Treponema pallidum*
Tetanus	*Clostridium tetani*
Bacterial dysentery	*Shigella dysenteriae*
Gonorrhoea	*Neisseria gonorrhoeae*
Diseases of animals	
Anthrax	*Bacillus anthracis*
Pneumonia	*Diplococcus pneumoniae*
Black leg of cattle	*Clostridium chauvei*
Diseases of plants	
Crown galls	*Agrobacterium tumefaciens*
Pear blight	*Pseudomonas amylovorus*
Ring rot of potato	*Corynebacterium sependonicum*
Citrus canker	*Xanthomonas citri*
Black rot of cabbage	*X. campestris*

symbiotic nitrogen-fixing bacteria. In addition, ***nitrifying bacteria*** (convert ammonia to nitrates) and ***ammonifying bacteria*** (convert amino acids to ammonia) also contribute to agricultural practices.

CONTROL OF PATHOGENIC BACTERIA

As already stated above, bacteria cause several diseases. There are different approaches to control pathogenic bacteria

- Use of antibiotics
- Treatment with antiseptics and disinfectants.
- Sterilization and autoclaving
- Radiation treatment
- Biological control.

In addition to the above, vaccines have been developed against certain pathogens to control diseases in humans and animals.

FUNGI

Mycology, a branch of microbiology, deals with the study of fungi. Fungi are a group of eukaryotic, heterotrophic organisms. They are dependent on organic compounds for their nutrition. In general, fungi can withstand extreme environmental conditions better than most of the microorganisms.

CHARACTERISTICS OF FUNGI

Fungi cells are usually larger than the bacteria. The sizes may range 1-5 µm in width and 5–35 µm in length. Fungal cells may be elongated or spherical.

The fungi are heterotrophic, since they cannot synthesize their own food from the inorganic compounds.

IMPORTANCE OF FUNGI

As is the case with bacteria, fungi are both friends and enemies of humans.

Beneficial aspects of fungi

The ***yeasts are useful*** for the following

- Alcohol fermentation e.g. *Saccharomyces cerevisiae*
- Production of vitamins e.g. *Ashbya gossypii.*
- Citric acid fermentation e.g. *Candida* sp.
- Baker's yeast e.g. *S. cerevisiae.*

The ***application of molds*** are listed

- Production of enzymes e.g. *Aspergillus* sp.
- Citric acid fermentation e.g. *Aspergillus niger.*
- Penicillin production e.g. *Penicillium notatum.*
- Steroid transformation e.g. *Rhizopus* sp.
- Gluconic acid production e.g. *Aspergillus niger.*

Harmful aspects of fungi

- Spoilage of foods e.g. moldy bread, rot of fruits and vegetables.
- Deterioration of textiles made up of cotton.
- Damage to paper.
- Diseases caused by fungi e.g. ring worm of the scalp in children caused by *Microsporum audouinii.*

CONTROL OF FUNGI

The fungal growth can be controlled by using phenol and its derivatives e.g. cresol, ethylphenol, propylphenol, butylphenol. Chlorine and chlorine compounds are also useful in this regard.

Vegetative cells of yeasts and other fungi can be destroyed by moist heat at 50–60°C for about 5–10 minutes. Spores, however require higher temperature 20–80°C.

VIRUSES

A virus may be regarded as a small microorganism with a nucleoprotein entity which lives and multiplies in the living cells of other organisms. Thus, viruses as are unable to live in a cell-free environment, and become active and multiply as they enter living cells.

CHARACTERISTICS OF VIRUSES

Viruses are the smallest living entities. They do not possess the usual cellular structures. Viruses are composed of either DNA or RNA, and not both of them together. ***Retroviruses*** is the name given to viruses containing RNA as the genetic material.

Viruses do not possess independent metabolisms, and further they also lack respiratory machinery.

BIOLOGICAL STATUS OF VIRUSES

Organismal theory : Viruses are considered as primitive living microorganisms as they possess genetic material, biosynthetic machinery and certain enzymes, besides being infective.

TABLE 63.2 A selected list of common viral diseases

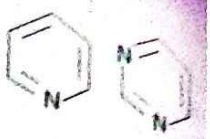
Diseases of humans	*Virus*
Common cold	*Rhinoviruses*
Small pox	*Pox virus*
Rabies	*Rabies virus*
Infectious hepatitis	*Hepatitis virus*
Measles	*Measles virus*
Poliomyelitis	*Polio virus*
Influenza	*Influenza virus*
Encaphalitis	*Encephalitis virus*
Mumps	*Mumps virus*
Diseases of plants	
Tobacco mosaic	*Tobacco mosaic virus*
Potato leaf roll	*Potato leaf roll virus*
Tomato leaf curl	*Tomato leaf curl virus*

Molecular theory : Some people consider viruses are non-living chemical molecules rather than living beings. They argue that viruses do not possess any cellular structures, cannot respire and cannot live independently.

IMPORTANCE OF VIRUSES

Viruses are associated with several diseases in humans and plants (***Table 63.2***). Besides the common diseases such as common cold, and rabies, viruses cause diseases such as AIDS, i.e. human immunodeficiency virus (HIV) is responsible for AIDS, the dreaded and incurable disease.

64 Bioorganic and Biophysical Chemistry, and Biochemistry Tools



The fundamental principles of bioorganic and biophysical chemistry, besides the common tools employed in biochemistry are briefly described in this chapter.

INTRODUCTION TO BIOORGANIC CHEMISTRY

As life comes from previous life, it was believed for a long that the carbon compounds of organisms (hence the name organic) arose from life only. This is referred to as ***vital force theory***. Friedrich Wohler (1825) first discovered that urea (NH_2—CO—NH_2), the organic compound, could be prepared by heating ammonium cyanate (NH_4NCO), in the laboratory. Thereafter, thousands and thoushands of organic compounds have been synthesized outside the living system.

Organic chemistry broadly deals with the chemistry of carbon compounds, regardless of their origin. Biochemistry, however, is concerned with the ***carbon chemistry of life only***. The general principles of organic chemistry provide strong foundations for understanding biochemistry. However, biochemistry exclusively deals with the reactions that occur in the living system in aqueous medium.

MOST COMMON ORGANIC COMPOUNDS FOUND IN LIVING SYSTEM

The organic compounds, namely carbohydrates, lipids, proteins, nucleic acids and vitamins are the most common organic compounds of life.

COMMON FUNCTIONAL GROUPS IN BIOCHEMISTRY

Most of the physical and chemical properties of organic compounds are determined by their functional groups. Biomolecules possess certain functional groups which are their reactive centres. The common functional groups of importance in biomolecules are presented in ***Table 64.1***.

COMMON RING STRUCTURES IN BIOCHEMISTRY

There are many homocyclic and heterocyclic rings, commonly encountered in biomolecules. A selected list of them is given in ***Fig. 64.1***.

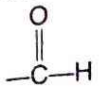
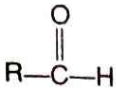
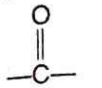
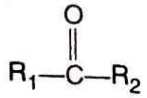
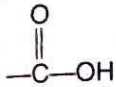
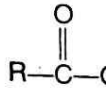
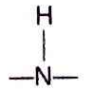
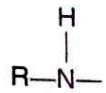
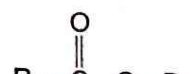
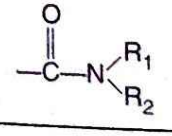
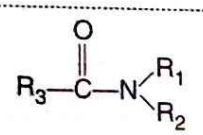
Homocyclic rings

Phenyl ring derived from benzene is found in several biomolecules (phenylalanine, tyrosine, catecholamines). Phenanthrene and cyclopentane form the backbone of steroids (cholesterol, aldosterone). Coenzyme Q is an example of benzoquinone while vitamin K is a naphthoquinone.

Heterocyclic rings

Furan is the ring structure found in pentoses. Pyrrole is the basic unit of porphyrins found in many biomolecules (heme) while pyrolidine is the ring present in the amino acid, proline. Thiophen ring is a part of the vitamin biotin. The amino acid histidine contains imidazole.

Table 64.1 Common functional groups of importance in biomolecules

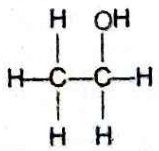
Functional group Name	Group	General structural formula	Type of compound	Examples of biomolecule(s)
Hydroxyl	—OH	R—OH	Alcohol	Glycerol, ethanol
Aldehyde			Aldehyde	Glyceraldehyde, glucose
Keto			Ketone	Fructose, sedoheptulose
Carboxyl			Carboxylic acid	Acetic acid, palmitic acid
Amino	$-NH_2$	$R-NH_2$	Amino acid	Alanine, serine
Imino			Imino acid	Proline, hydroxyproline
Sulfhydryl	—SH	R—SH	Thiol	Cysteine, coenzyme A
Ether	—O—	$R_1—O—R_2$	Ether	Thromboxane A_2
Ester			Ester	Cholesterol ester
Amido			Amide	N-Acetylglucosamine

Pyran structure is found in hexoses. Pyridine nucleus is a part of the vitamins-niacin and pyridoxine. Pyrimidines (cytosine, thymine) and purines (adenine, guanine) are the constituents of nucleotides and nucleic acids. Indole ring is found in the amino acid tryptophan. Purine and indole are examples of fused heterocyclic rings.

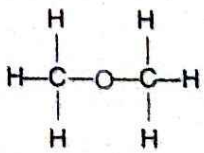
ISOMERISM

The compounds possessing identical molecular formulae but different structures are referred to as **isomers.** The phenomenon of existence of isomers is called isomerism (Greek : isos—equal; meros—parts). Isomers differ from each other in physical and chemical properties. ***Isomerism is partly responsible for the existence of a large number of organic molecules.***

Consider the molecular formula—C_2H_6O. There are two important isomers of this — ethyl alcohol (C_2H_5OH) and diethyl ether (CH_3OCH_3) as shown below :



Ethyl alcohol



Diethyl ether

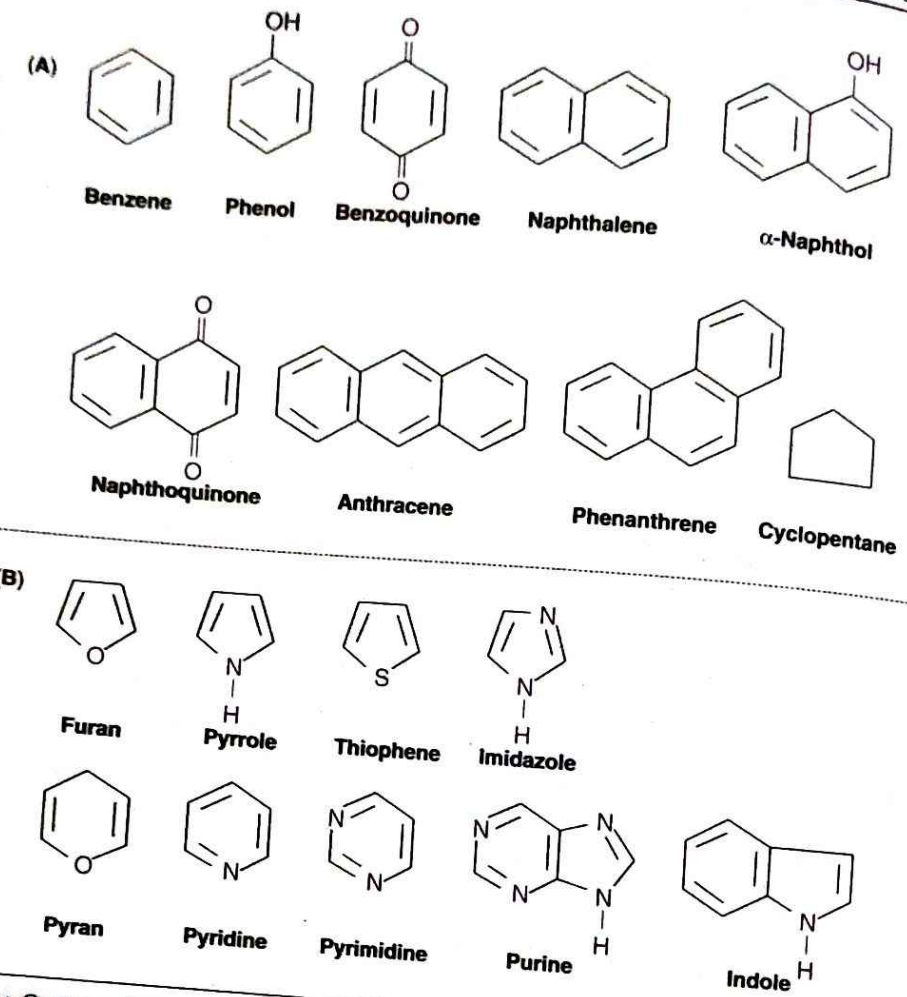


Fig. 64.1 : Common ring structures found in biomolecules (A) Homocyclic rings (B) Heterocyclic rings.

Isomerism is broadly divided into two categories — structural isomerism and stereo-isomerism.

Structural isomerism

The difference in the arrangement of the atoms in the molecule (i.e. molecular framework) is responsible for structural isomerism. This may be due to variation in carbon chains ***(chain isomerism)*** or difference in the position of functional groups ***(position isomerism)*** or difference in both molecular chains and functional groups ***(functional isomerism)***.

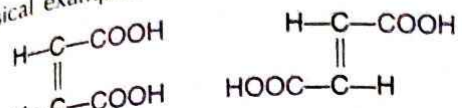
Structural isomerism, as such, is more common in general organic molecules. ***Tautomerism,*** a type of structural isomerism, occurs due to the migration of an atom or group from one position to the other e.g. purines and pyrimidines.

Stereoisomerism

Stereoisomerism (*Greek* : stereos—space occupying) is, perhaps, more relevant and important to biomolecules. The differential space arrangement of atoms or groups in molecules gives rise to stereoisomerism. Thus, stereoisomers have the same structural formula but differ in their spacial arrangement.

Stereoisomerism is of two types—***geometric isomerism*** and ***optical isomerism.***

Geometrical isomerism : This is also called *cis-trans* isomerism and is exhibited by certain molecules possessing double bonds. Geometrical isomerism is due to restriction of freedom of rotation of groups around a carbon-carbon double bond (C = C). Maleic acid and fumaric acid are classical examples of *cis-trans* isomerism.



Maleic acid *(cis)* **Fumaric acid** *(trans)*

When similar groups lie on the same side, it is called *cis* isomer (*Latin* : cis—on the same side). On the other hand, when similar groups lie on the opposite sides, it is referred to as *trans* isomer (*Latin* : trans—across). As is observed from the above structure, maleic acid is a *cis* form while fumaric acid is a *trans* form.

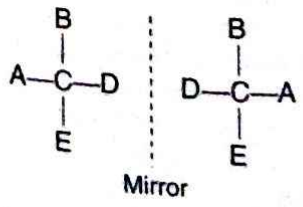
Geometric isomerism is also observed in sterols and porphyrins. *cis-trans* isomers differ in physical and chemical properties.

Optical isomerism : Optical isomers or enantiomers occur due to the presence of an ***asymmetric carbon*** (a chiral carbon). Optical isomers differ from each other in their optical activity to rotate the plane of polarized light.

What is an asymmetric carbon?

An object is said to be symmetrical if it can be divided into equal halves e.g. a ball. Objects which cannot be divided into equal halves are asymmetric, e.g. hand. An asymmetric object cannot coincide with its mirror image. For instance, left hand is the mirror image of right hand and these two can never be superimposed. In contrast, a symmetrical object like a ball superimposes its image.

A carbon is said to be ***chiral*** (*Greek* : hand) or asymmetric when it is attached to four different groups. Their mirror images do not superimpose with each other.



The number of possible optical isomers of a molecule depends upon the specific number of chiral carbon (n). It is given by 2^n.

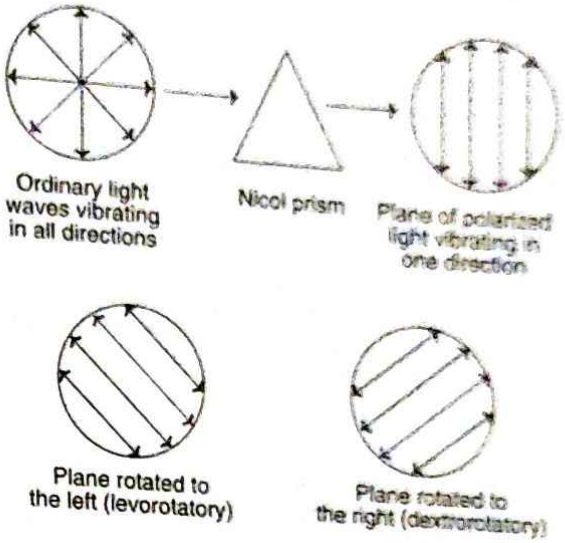


Fig. 64.2 : Diagrammatic illustration of optical activity

What is optical activity?

The ordinary light propagates in all directions. However, on passing ordinary light through a Nicol prism, the plane of polarized light vibrates in one direction only (***Fig. 64.2***).

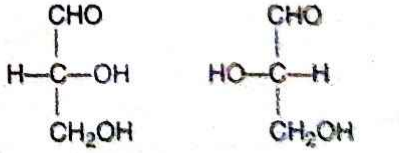
Certain organic compounds (optical isomers) which are said to exhibit optical activity rotate the plane of polarized light either to the left or to the right.

The term ***levorotatory*** (indicated by l or (–) sign) is used for the substances which rotate the plane of polarized light to the left. On the other hand, the term ***dextrorotatory*** (indicated by d or (+) sign) is used for substances rotating the plane of polarized light to right (***Fig. 64.2***).

The term ***racemic*** mixture represents equal concentration of d and l forms which cannot rotate the plane of polarized light.

Configuration of chiral molecules

While representing the configuration of chiral molecules, the configuration of ***glyceraldehyde*** is taken as a ***reference standard***.



D-Glyceraldehyde **L-Glyceraldehyde**

It must, however, be remembered that D- and L- do not represent the direction of the rotation of plane of polarized light.

Existence of chiral biomolecules

As you know, you can never come across anybody who is your mirror image. The same is true with biomolecules. Only one type of molecules (D or L) are found in the living system. Thus, the naturally occurring amino acids are of L-type while the carbohydrates are of D-type.

OVERVIEW OF BIOPHYSICAL CHEMISTRY

The general laws and principles of chemistry and physics are applicable to biochemistry as well. It is, therefore, worthwhile to have a brief understanding of some of the basic chemical and physical principles that have direct relevance to life.

It must, however, be remembered that all these aspects are quite unrelated to each other.

ACIDS AND BASES

According to Lowry and Bronsted, an acid is defined as a substance that gives off protons while a base is a substance that accepts protons. Thus, ***an acid is a proton (H^+) donor and a base is a proton acceptor.***

Alkalies : The metallic hydroxides such as NaOH and KOH are commonly referred to as alkalies. These compounds do not directly satisfy the criteria of bases. However, they dissociate to form metallic ion and OH^- ion. The latter, being a base, accepts H^+ ions.

Ampholytes : The substances which can function both as acids and bases are referred to as ampholytes. Water is the best example for ampholytes.

Hydrogen ion concentration (pH)

The acidic or basic nature of a solution is measured by H^+ ion concentration. The strength of H^+ ions in the biological fluids is exceedingly low. For this reason, the conventional units such as moles/l or g/l are not commonly used to express H^+ ion concentration.

Sorenson (1909) introduced the term pH to express H^+ ion concentration. ***pH is defined as the negative logarithm of H^+ ion concentration.***

$$pH = -\log [H^+]$$

BUFFERS

The pH of a given solution can be easily altered by the addition of acids or bases. ***Buffers are defined as the solutions which resist change in pH by the addition of small amounts of acids or bases.*** A buffer usually consists of a weak acid and its salt (e.g. acetic acid and sodium acetate) or a weak base and its salt (e.g. ammonium hydroxide and ammonium chloride). Several buffers can be prepared in the laboratory. Nature has provided many buffers in the living system.

Mechanism of buffer action

Let us consider the buffer pair of acetic acid and sodium acetate. Acetic acid, being a weak acid, feebly ionizes. On the other hand, sodium acetate ionizes to a large extent.

$$CH_3COOH \rightleftharpoons CH_3COO^- + H^+$$

$$CH_3COONa \rightleftharpoons CH_3COO^- + Na^+$$

When an acid (say HCl) is added, the acetate ions of the buffer bind with H^+ ions (of HCl) to form acetic acid which is weakly ionizing. Therefore, the pH change due to acid is resisted by the buffer.

$$H^+ + CH_3COO^- \longrightarrow CH_3COOH$$

When a base (say NaOH) is added the H^+ ions of the buffer (acetic acid) combine with OH^- ions to form water, which is weakly dissociated. Thus, the pH change due to base addition is also prevented by the buffer.

$$OH^- + H^+ \longrightarrow H_2O$$

Buffering capacity : The efficiency of a buffer in maintaining a constant pH on the addition of acid or base is referred to as buffering capacity. It mostly depends on the concentration of the buffer components.

SOLUTIONS

Solutions may be regarded as mixtures of substances. In general, a solution is composed of two parts — ***solute*** and ***solvent.*** The substance that

is dissolved is solute and the medium that dissolves the solute is referred to as solvent. The particle size of a solute in solution is < 1 nm.

The relative concentrations of substances in a solution can be measured by several ways.

Per cent concentration : This represents parts per 100. The most frequently used is weight per volume (w/v) e.g. 9% saline (9 g/100 ml solution).

Parts per million (ppm) : This refers to the number of parts of a substance in one million parts of the solution. Thus 10 ppm chlorine means 10 µg of chlorine in 1 g of water.

Molarity (M) : It is defined as the number of moles of solute per liter solution. NaCl has a molecular weight of 58.5. To get one molar (1 M) or one mole solution of NaCl, one gram molecular weight (58.5 g) of it should be dissolved in the solvent (H_2O) to make to a final total volume of 1 liter. For smaller concentrations, millimole and micromole are used.

Molality : It represents the number of moles of solute per 1,000 g of solvent. One molal solution can be prepared by dissolving 1 mole of solute in 1,000 g of solvent.

Normality : Molarity is based on molecular weight while normality is based on equivalent weight. One gram equivalent weight of an element or compound represents its capacity to combine or replace 1 mole of hydrogen.

COLLOIDAL STATE

Thomas Graham (1861), regarded as the 'father of colloidal chemistry', divided substances into two classes—***crystalloids*** and ***colloids.***

Crystalloids are the substances which in solution can freely pass (diffuse) through parchment membrane e.g. sugar, urea, NaCl. Colloids (*Greek* : glue-like), on other hand, are the substances that are retained by parchment membrane e.g. gum, gelatin, albumin. The above classification of Graham is no longer tenable, since any substance can be converted into a colloid by suitable means. For instance, sodium chloride in benzene forms a colloid.

Colloidal state : As such, there are no group of substances as colloids, rather, substances can exist in the form of ***colloidal state*** or ***colloidal system.*** Colloidal state is characterized by the particle size of 1 to 100 nm. When the particle size is <1 nm, it is in true solution. For the particle sizes >100 nm, the matter exists as a visible precipitate. Thus, the colloidal state is an intermediate between true solution and precipitate.

Phases of colloids : Colloidal state is heterogeneous with two phases.

1. **Dispersed phase** (internal phase) which constitutes the colloidal particles.
2. **Dispersion medium** (external phase) which refers to the medium in which the colloidal particles are suspended.

Classification of colloids

Based on the affinity of dispersion medium with dispersed phase, colloids are classified as ***lyophobic*** and ***lyophilic*** colloids.

1. **Lyophobic** (*Greek* : solvent-hating) **:** These colloids do not have any attraction towards dispersion medium. When water is used as dispersion medium, the colloids are referred to as ***hydrophobic.***

2. **Lyophilic** (*Greek* : solvent-loving) **:** These colloids have distinct affinity towards dispersion medium. The term ***hydrophilic*** is used for the colloids when water is the dispersion medium.

Biological importance of colloids

1. **Biological fluids as colloids :** These include blood, milk and cerebrospinal fluid.

2. **Biological compounds as colloidal particles :** The complex molecules of life, the high molecular weight proteins, complex lipids and polysaccharides exist in colloidal state.

3. **Fat digestion and absorption :** The formation of emulsions, facilitated by the emulsifying agents bile salts, promotes fat digestion and absorption in the intestinal tract.

4. **Formation of urine :** The filtration of urine is based on the principle of dialysis.

DIFFUSION

The molecules in liquids or gases are in continuous motion. ***Diffusion may be regarded as the movement of solute molecules from a higher concentration to a lower concentration.*** Diffusion

is more rapid in gases than in liquids. The smaller particles diffuse faster than the larger ones. The greater the temperature, the higher is the rate of diffusion.

Applications of diffusion

1. Exchange of O_2 and CO_2 in lungs and in tissues occurs through diffusion.

2. Certain nutrients are absorbed by diffusion in the gastrointestinal tract e.g. pentoses, minerals, water soluble vitamins.

3. Passage of the waste products namely ammonia, in the renal tubules occurs due to diffusion.

OSMOSIS

Osmosis (*Greek* : push) refers to the ***movement of solvent*** (most frequently water) ***through a semipermeable membrane.***

The flow of solvent occurs from a solution of low concentration to a solution of high concentration, when both are separated by a semipermeable membrane.

Osmotic pressure

Osmotic pressure may be defined as the excess pressure that must be applied to a solution to prevent the passage of solvent into the solution, when both are separated by a semipermeable membrane.

The solutions that exert the same osmotic pressure are said to be ***isoosmotic.*** The term ***isotonic*** is used when a cell is in direct contact with an isoosmotic solution (0.9% NaCl) which does not change the cell volume and, thus, the cell tone is maintained. A solution with relatively greater osmotic pressure is referred to as ***hypertonic***. On the other hand, a solution with relatively lower pressure is ***hypotonic.***

Units of osmotic pressure : Osmotic pressure of biological fluids is frequently expressed as ***milliosmoles.*** The osmotic pressure of plasma (of human blood) is 280-300 milliosmoles/l.

Applications of osmosis

1. **Fluid balance and blood volume :** The fluid balance of the different compartments of the body is maintained due to osmosis.

2. **Red blood cells and fragility :** When the RBC are kept in ***hypotonic solution*** (say 0.4% NaCl), the cells bulge due to entry of water which often causes rupture of plasma membrane of RBC ***(hemolysis)***.

3. **Transfusion :** Isotonic solutions of NaCl (0.9%) or glucose (5%) or a suitable combination of these two are commonly used in transfusion in hospitals for the treatment of dehydration, burns etc.

4. **Action of purgatives :** The mechanism of action of purgatives is mainly due to osmotic phenomenon. For instance, epson ($MgSO_4$ $7H_2O$) or Glauber's (Na_2SO_4 $10H_2O$) salts withdraw water from the body, besides preventing the intestinal water absorption.

5. **Edema due to hypoalbuminemia :** Disorders such as kwashiorkor and glomerulonephritis are associated with lowered plasma albumin concentration and edema. Edema is caused by reduced oncotic pressure of plasma, leading to the accumulation of excess fluid in tissue spaces.

VISCOSITY

Liquid or fluid has a tendency to flow which is referred to as fluidity. The term viscosity may be defined as the ***internal resistance offered by a liquid or a gas to flow***. The property of viscosity is due to frictional forces between the layers while their movement occurs.

Viscosity of colloidal solutions, particularly lyophilic colloids, is generally higher than true solutions.

Units of viscosity : The unit of viscosity is ***poise***, after the scientist Poiseuille, who first systematically studied the flow of liquids. A poise represents dynes/cm^2.

Applications of viscosity

1. **Viscosity of blood :** Blood is about 4 times more viscous than water. The viscosity of blood is mainly attributed to suspended blood cells and colloidal plasma proteins.

2. **Viscosity change in muscle :** Excitation of the muscle is associated with increase in the viscosity of the muscle fibres. This delays the change in the tension of the contracting muscle.

SURFACE TENSION

A molecule in the interior of a liquid is attracted by other molecules in all directions. In contrast, a molecule on the surface is attracted only downwards and sideways and not upwards. Due to this, the surface layer behaves like a stretched film. ***Surface tension is the force with which the molecules on the surface are held together.*** It is expressed as dynes/cm. Surface tension decreases with increase in temperature.

Applications of surface tension

1. **Digestion and absorption of fat :** Bile salts reduce the surface tension. They act as detergents and cause emulsification of fat, thereby allowing the formation of minute particles for effective digestion and absorption.

2. **Surfactants and lung function :** The low surface tension of the alveoli keeps them apart and allows an efficient exchange of gases in lungs. In fact, certain surfactants, predominantly dipalmitoyl phosphatidyl choline (dipalmitoyl lecithin), are responsible for maintaining low surface tension in the alveoli. Surfactant deficiency causes respiratory distress syndrome in the infants.

ADSORPTION

Adsorption is a surface phenomenon. It is the process of accumulation of a substance (adsorbate) on the surface of another substance (adsorbent). Adsorption differs from absorption, as the latter involves the diffusion into the interior of the material.

Applications of adsorption

1. **Formation of enzyme-substrate complex :** For the catalysis to occur in biological system, formation of enzyme-substrate complex is a prerequisite. This happens by adsorption of substrate on the enzyme.

2. **Action of drugs and poisons :** On adsorption at the cell surface, drugs and poisons exert their action.

3. **Adsorption in analytical biochemistry :** The principle of adsorption is widely employed in the chromatography technique for the separation and purification of compounds (enzymes, immuno-globulins).

ISOTOPES

Isotopes are defined as the elements with same atomic number but different atomic weights. They possess the same number of protons but differ in the neutrons in their nuclei. Therefore, isotopes (Greek iso—equal; tope—place) occupy the same place in the periodic table. The chemical properties of different isotopes of a particular element are identical.

Isotopes are of two types—***stable*** and ***unstable***. The latter are more commonly referred to as radioactive isotopes and they are of particular interest to biochemists.

Stable isotopes

They are naturally occurring and do not emit radiations (non-radioactive) e.g. deuterium (heavy hydrogen) ^{2}H; ^{13}C; ^{15}N; ^{18}O. Stable isotopes can be identified and quantitated by mass spectrometry or nuclear magnetic resonance (NMR). They are less frequently used in biochemical investigations.

Radioactive isotopes

The atomic nucleus of radioactive isotopes is unstable and, therefore, undergoes decay. The radioactive decay gives rise to one of the following 3 ionizing radiations.

1. **α-Rays** — an α particle possessing 2 protons i.e. helium nuclei.

2. **β-Rays** — due to the emission of electrons.

3. **γ-Rays** — due to emission of high energy photons.

The β and γ emitting radioisotopes are employed in biochemical research.

Applications of radioisotopes in biochemistry

Radioactive isotopes have become indispensable tools of biochemistry. They can be conveniently used as tracers in biochemical research since the chemical properties of different isotopes of a particular element are identical. Therefore, the living cells cannot distinguish the radioactive isotope from a normal atom. A few important application of radioisotopes are listed

1. By the use of isotope tracers, the metabolic origin of complex molecules such as heme, cholesterol, purines and phospholipids can be determined.

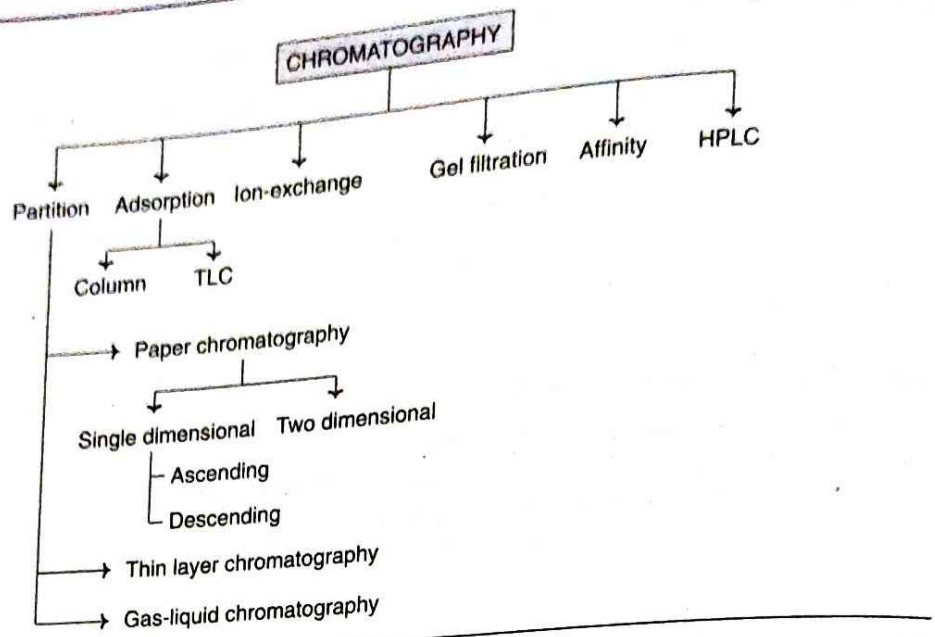


***Fig. 64.3** : Important types of chromatography*
(HPLC—High performance liquid chromatography; TLC—Thin layer chromatography)

2. The precursor-product relationship in several metabolic pathways has been investigated by radioisotopes. e.g. Krebs cycle, β-oxidation of fatty acids, urea cycle, fatty acid synthesis.

3. Radioisotopes are conveniently used in the study of metabolic pools (e.g. amino acid pool) and metabolic turnovers (e.g. protein turnover).

4. Certain endocrine and immunological studies also depend on the use of radioisotopes e.g. radioimmunoassay.

5. Radioisotopes are employed in elucidating drug metabolism.

TOOLS OF BIOCHEMISTRY

The foundations for the present (and the future, of course!) knowledge of biochemistry are based on the laboratory tools employed for biochemical experimentation. The basic principles of some of the commonly employed tools are described here.

CHROMATOGRAPHY

Chromatography is one of the most useful and popular tools of biochemistry. It is an analytical technique dealing with the ***separation of closely related compounds from a mixture***. These include proteins, peptides, amino acids, lipids, carbohydrates, vitamins and drugs.

Principles and classification

Chromatography (*Greek* : chroma — colour; graphein — to write), usually consists of a ***mobile phase*** and a ***stationary phase.*** The mobile phase refers to the mixture of substances (to be separated), dissoved in a liquid or a gas. The stationary phase is a porous solid matrix through which the sample contained in the mobile phase percolates. The interaction between the mobile and stationary phases results in the separation of the compounds from the mixture. These interactions include the physico-chemical principles such as ***adsorption, partition, ion-exchange, molecular sieving*** and ***affinity.***

The interaction between stationary phase and mobile phase is often employed in the classification chromatography e.g. partition, adsorption, ion-exchange. Further, the classification of chromatography is also based either on the nature of the stationary phase (paper, thin layer, column), or on the nature of both mobile and stationary phases (gas-liquid chromatography). A summary of the different methods (classes) of chromatography is given in ***Fig. 64.3.***

1. **Partition chromatography** : The molecules of a mixture get partitioned between the stationary phase and mobile phase depending on their relative affinity to each one of the phases.

2. **Adsorption column chromatography** : The adsorbents such as silica gel, alumina, charcoal powder and calcium hydroxyapatite are packed into a column in a glass tube. This serves as the stationary phase. The sample mixture in a solvent is loaded on this column. The individual components get differentially adsorbed on to the adsorbent. The elution is carried out by a buffer system (mobile phase). The individual compounds come out of the column at different rates which may be separately collected and identified. For instance, amino acids can be identified by ninhydrin colorimetric method. An automated column chromatography apparatus — ***fraction collector*** — is frequently used nowadays.

3. **High performance liquid chromatography (HPLC)** : In general, the chromatographic techniques are slow and time consuming. The separation can be greatly improved by applying high pressure in the range of 5,000-10,000 psi (pounds per square inch), hence this technique is also referred (less frequently) to as high pressure liquid chromatography.

More information on chromatography with special reference to ***gel-filtration chromatography, ion-exchange chromatography, affinity chromatography and hydrophobic interaction chromatography*** is given under downstream processing (Chapter 20).

ELECTROPHORESIS

The ***movement of charged particles*** (ions) ***in an electric field*** resulting in their migration towards the oppositely charged electrode is known as electrophoresis. Molecules with a net positive charge (cations) move towards the negative cathode while those with net negative charge (anions) migrate towards positive anode. Electrophoresis is a widely used analytical technique for the separation of biological molecules such as plasma proteins, lipoproteins and immunoglobulins.

Different types of electrophoresis

1. **Zone electrophoresis** : An inert supporting material such as paper or gel are used.

2. **Isoelectric focussing** : This technique is primarily based on the ***immobilization of the molecules at isoelectric pH*** during electrophoresis. Stable pH gradients are set up (usually in a gel) covering the pH range to include the isoelectric points of the components in a mixture. Isoelectric focussing can be conveniently used for the purification of proteins.

3. **Immunoelectrophoresis** : This technique involves combination of the principles of electrophoresis and immunological reactions. Immunoelectrophoresis is useful for the analysis of complex mixtures of antigens and antibodies.

PHOTOMETRY—COLORIMETER AND SPECTROPHOTOMETER

Photometry broadly deals with the study of the phenomenon of light absorption by molecules in solution. The specificity of a compound to absorb light at a particular wavelength (monochromatic light) is exploited in the laboratory for quantitative measurements. From the biochemist's perspective, photometry forms an important laboratory tool for accurate estimation of a wide variety of compounds in biological samples. Colorimeter and spectrophotometer are the laboratory instruments used for this purpose. They work on the principles discussed below.

When a light at a particular wavelength is passed through a solution (incident light), some amount of it is absorbed and, therefore, the light that comes out (transmitted light) is diminished. The nature of light absorption in a solution is governed by ***Beer-Lambert law.***

Beer's law states that the amount of transmitted light decreases exponentially with an increase in the concentration of absorbing material (i.e. the amount of light absorbed depends on the concentration of the absorbing molecules). And according to Lambert's law, the transmitted light decreases exponentially with increase in the thickness of the absorbing molecules (i.e. the amount of light absorbed is dependent on the thickness of the medium).

The ratio of transmitted light (I) to that of incident light (I_0) is referred to as transmittance (T).

$$T = \frac{I}{I_0}$$

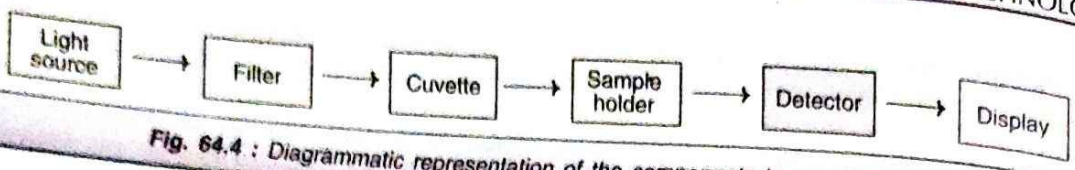


Fig. 64.4 : Diagrammatic representation of the components in a colorimeter.

Absorbance (A) or optical density (OD) is very commonly used in laboratories. The relation between absorbance and transmittance is expressed by the following equation.

$$A = 2 - \log_{10} T$$

$$= 2 - \log\% \ T$$

Colorimeter

Colorimeter (or photoelectric colorimeter) is the instrument used for the measurement of coloured substances. This instrument is operative in the visible range (400-800 nm) of the electromagnetic spectrum of light. The working of colorimeter is based on the principle of Beer-Lambert law.

The colorimeter, in general consists of light source, filter sample holder and detector with display (meter or digital). A filament lamp usually serves as a light source. The filters allow the passage of a small range of wave length as incident light. The sample holder is a special glass cuvette with a fixed thickness. The photoelectric selenium cells are the most common detectors used in colorimeter. The diagrammatic representation of a colorimeter is depicted in ***Fig. 64.4***.

Spectrophotometer

The spectrophotometer primarily differs from colorimeter by covering the ultraviolet region (200-400 nm) of the electromagnetic spectrum. Further, the spectrophotometer is more sophisticated with several additional devices that ultimately increase the sensitivity of its operation severalfold when compared to a colorimeter. A precisely selected wavelength (say 234 nm or 610 nm) in both ultra violet and visible range can be used for measurements. In place of glass cuvettes (in colorimeter), quartz cells are used in a spectrophotometer.

The spectrophotometer has similar basic components as described for a colorimeter (***Fig. 64.4***) and its operation is also based on the Beer-Lambert law.

ULTRACENTRIFUGATION

Ultracentrifugation is an indispensable tool for the isolation of subcellular organelles, proteins and nucleic acids. In addition, this technique is also employed for the determination of molecular weights of macromolecules.

The rate at which the sedimentation occurs in ultracentrifugation primarily depends on the size and shape of the particles or macromolecules (i.e. on the molecular weight). It is expressed in terms of ***sedimentation coefficient***(s).

The sedimentation coefficient has the units of seconds. It was usually expressed in units of 10^{-13} s (since several biological macromolecules occur in this range), which is designated as one ***Svedberg unit.*** For instance, the sedimentation coefficient of hemoglobin is 4×10^{-13} s or 4S; ribonuclease is 2×10^{-13} s or 2S. Conventionally, the subcellular organelles are often referred to by their S value e.g. 70S ribosome.

Isolation of subcellular organelles by centrifugation

The cells are subjected to disruption by sonication or osmotic shock or by use of homogenizer. This is usually carried out in an isotonic (0.25 M) sucrose. The advantage with sucrose medium is that it does not cause the organelles to swell. The subcellular particles can be separated by differential centrifugation. The most commonly employed laboratory method separates subcellular organelles into 3 major fractions—nuclear, mitochondrial and microsomal (***Fig. 64.5***).

When the homogenate is centrifuged at 700 g for about 10 min, the nuclear fraction (includes plasma membrane) gets sedimented. On centrifuging the supernatant (I) at 15,000 g for about 5 min mitochondrial fraction (that includes lysosomes, peroxisomes) is pelleted. Further centrifugation of the supernatant (II) at 100,000 g for about 60 min separates microsomal fraction (that includes ribosomes and endoplasmic reticulum). The supernatant (III) then obtained corresponds to the cytosol.

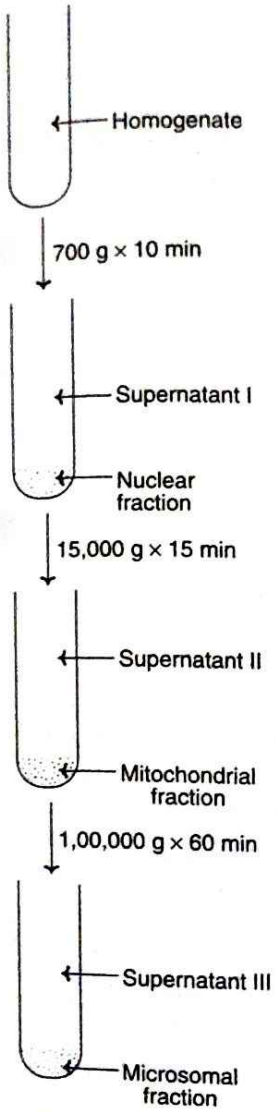


Fig. 64.5 : Separation of subcellular fractions by differential centrifugation.

The purity (or contamination) of the subcellular fractionation can be checked by the use of ***marker enzymes.*** DNA polymerase is the marker enzyme for nucleus, while glutamate dehydrogenase and glucose 6-phosphatase are the markers for mitochondria and ribosomes, respectively. Hexokinase is the marker enzyme for cytosol.

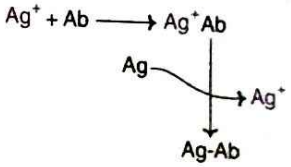
RADIOIMMUNOASSAY

Radioimmunoassay (RIA) was developed in 1959 by Solomon, Benson and Rosalyn Yalow for the estimation of insulin in human serum. This technique has revolutionized the estimation of several compounds in biological fluids that are found in exceedingly low concentrations (nanogram or picogram). RIA is a highly sensitive and specific analytical tool.

Principle

Radioimmunoassay combines the principles of ***radioactivity*** of isotopes and ***immunological*** reactions of antigen and antibody, hence the name.

The principle of RIA is primarily based on the competition between the labeled and unlabeled antigens to bind with antibody to form antigen-antibody complexes (either labelled or unlabeled). The unlabeled antigen is the substance (say insulin) to be determined. The antibody to it is produced by injecting the antigen to a goat or a rabbit. The specific antibody (Ab) is then subjected to react with unlabeled antigen in the presence of excess amounts of isotopically labeled (^{131}I) antigen (Ag^*) with known radioactivity. There occurs a competition between the antigens (Ag^* and Ag) to bind the antibody. Certainly, the labeled Ag^* will have an upper hand due to its excess presence.



As the concentration of unlabeled antigen (Ag) increases the amount of labelled antigen-antibody complex (Ag^*-Ab) decreases. Thus, the concentration of Ag^*-Ab is inversely related to the concentration of unlabeled Ag i.e. the substance to be determined. This relation is almost linear. A standard curve can be drawn by using different concentrations of unlabeled antigen and the same quantities of antibody and labeled antigen.

The labeled antigen-antibody (Ag^*-Ab) complex is separated by precipitation. The radioactivity of ^{131}I present is Ag^*-Ab is determined.

Applications

RIA is no more limited to estimating of hormones and proteins that exhibit antigenic properties. By the use of ***haptens*** (small molecules such as dinitrophenol, which, by themselves, are not antigenic), several substances can be made

antigenic to elicit specific antibody responses. In this way, a wide variety of compounds have been brought under the net of RIA estimation. These include peptides, steroid hormones, vitamins, drugs, antibiotics, nucleic acids, structural proteins and hormone receptor proteins.

Radioimmunoassay has tremendous application in the diagnosis of hormonal disorders, cancers and therapeutic monitoring of drugs, besides being useful in biomedical research.

ENZYME-LINKED IMMUNOSORBANT ASSAY

Enzyme-linked immunosorbant assay (ELISA) is a non-isotopic immunoassay. An enzyme is used as a label in ELISA in place of radioactive isotope employed in RIA. ELISA is as sensitive as or even more sensitive than RIA. In addition, there is no risk of radiation hazards (as is the case with RIA) in ELISA.

Principle

ELISA is based on the immunochemical principles of antigen-antibody reaction. The stages of ELISA, depicted in ***Fig. 64.6***, are summarized.

1. The antibody against the protein to be determined is fixed on an inert solid such as polystyrene.

2. The biological sample containing the protein to be estimated is applied on the antibody coated surface.

3. The protein antibody complex is then reacted with a second protein specific antibody to which an enzyme is covalently linked. These enzymes must be easily assayable and produce preferably coloured products. Peroxidase, amylase and alkaline phosphatase are commonly used.

4. After washing the unbound antibody linked enzyme, the enzyme bound to the second antibody complex is assayed.

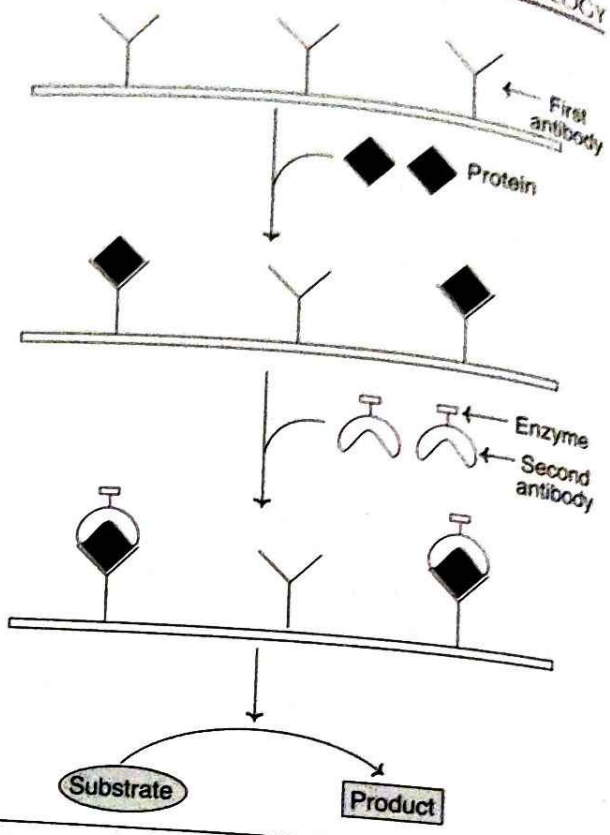


Fig. 64.6 : *Diagrammatic representation of enzyme-linked immunosorbant assay (ELISA).*

5. The enzyme activity is determined by its action on a substrate to form a product (usually coloured). This is related to the concentration of the protein being estimated.

Applications

ELISA is widely used for the determination of small quantities of proteins (hormones, antigens, antibodies) and other biological substances. The most commonly used pregnancy test for the detection of human chorionic gonadotropin (hCG) in urine is based on ELISA. By this test, pregnancy can be detected within few days after conception. ELISA is also useful for the diagnosis of AIDS.

acid, citrate. In the reactions that follow, the two carbons are oxidized to CO_2 and oxaloacetate is regenerated and recycled. ***Oxaloacetate is considered to play a catalytic role in citric acid cycle.*** The reactions of Krebs cycle are depicted in ***Fig. 67.5***.

GLUCONEOGENESIS

The synthesis of glucose or glycogen from non-carbohydrate compounds is known as gluconeogenesis. The major ***substrates***/precursors for gluconeogenesis are ***lactate***, ***pyruvate***, ***glucogenic amino acids***, ***propionate*** and ***glycerol.***

Location of gluconeogenesis

Gluconeogenesis occurs mainly in the ***cytosol***, although some precursors are produced in the mitochondria. Gluconeogenesis mostly takes place in liver and, to some extent, in kidney matrix (about one-tenth of liver capacity).

Reactions of gluconeogenesis

Gluconeogenesis closely resembles the reversed pathway of glycolysis, although it is not the complete reversal of glycolysis. Essentially, 3 (out of 10) reactions of glycolysis are irreversible. The seven reactions are common for both glycolysis and gluconeogenesis. The ***three irreversible steps*** of glycolysis are catalysed by the enzymes, namely ***hexokinase***, ***phosphofructokinase*** and ***pyruvate kinase***.

GLYCOGEN METABOLISM

Glycogen is the storage form of glucose in animals, as is starch in plants. It is stored mostly in liver (6-8%) and muscle (1-2%). Due to more muscle mass, the quantity of glycogen in muscle (250 g) is about three times higher than that in the liver (75 g).

Functions of glycogen

The prime function of liver glycogen is to maintain the blood glucose levels, particularly between meals. Liver glycogen stores increase in a well-fed state which are depleted during fasting. Muscle glycogen serves as a fuel reserve for the supply of ATP during muscle contraction.

GLYCOGENESIS

The ***synthesis of glycogen*** from glucose is glycogenesis. Glycogenesis takes place in the cytosol and requires ATP and UTP, besides glucose.

GLYCOGENOLYSIS

The ***degradation of stored glycogen*** in liver and muscle constitutes glycogenolysis. The pathway for the synthesis and degradation of glycogen are not reversible. An independent set of enzymes present in the cytosol carry out glycogenolysis. Glycogen is degraded by breaking α-1, 4- and α-1, 6-glycosidic bonds.

HEXOSE MONOPHOSPHATE SHUNT

Hexose monophosphate pathway or ***HMP shunt*** is also called ***pentose phosphate pathway*** or phosphogluconate pathway. ***This is an alternative pathway to glycolysis and TCA cycle for the oxidation of glucose.*** However, HMP shunt is more anabolic in nature, since it is concerned with the biosynthesis of NADPH and pentoses.

Location of the pathway

The enzymes of HMP shunt are located in the ***cytosol***. The tissues such as ***liver, adipose tissue, adrenal gland, erythrocytes, testes*** and ***lactating mammary gland***, are highly active in HMP shunt. Most of these tissues are involved in the biosynthesis of fatty acids and steroids which are dependent on the supply of NADPH.

Reactions of HMP shunt

The sequence of reactions of HMP shunt is depicted in ***Fig. 67.6***.

Significance of HMP shunt

HMP shunt is unique in generating two important products—***pentoses*** and ***NADPH***—needed for the biosynthetic reactions and other functions.

A. Importance of pentoses

In the HMP shunt, hexoses are converted into pentoses, the most important being ribose 5-phosphate. This pentose or its derivatives are useful for the ***synthesis of nucleic acids*** (RNA and DNA) and many ***nucleotides*** such as ATP, NAD^+, FAD and CoA.

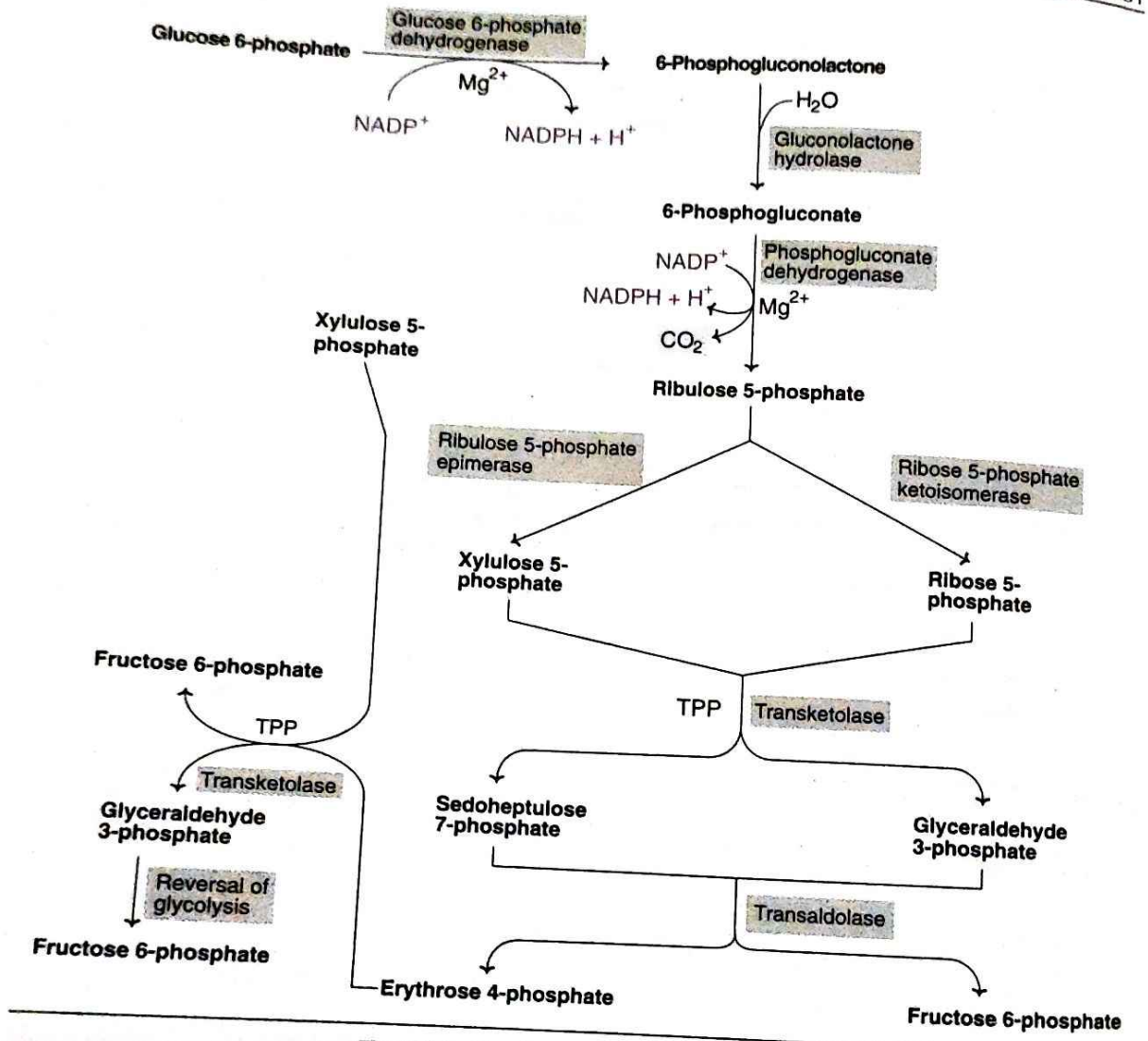


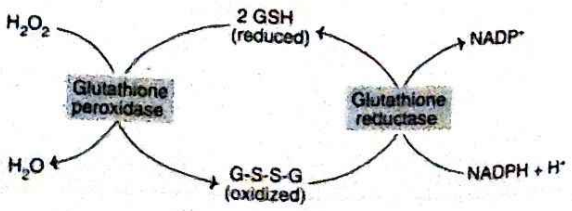
***Fig. 67.6** : The hexose monophosphate shunt.*

B. Importance of NADPH

1. NADPH is required for the reductive ***biosynthesis of fatty acids and steroids***, hence HMP shunt is more active in the tissues concerned with lipogenesis, e.g. adipose tissue, liver etc.

2. NADPH is used in the synthesis of certain amino acids involving the enzyme ***glutamate dehydrogenase.***

3. There is a continuous production of H_2O_2 in the living cells which can chemically damage unsaturated lipids, proteins and DNA. This is, however, prevented to a large extent through ***antioxidant reactions*** involving NADPH. Glutathione mediated reduction of H_2O_2 is given hereunder



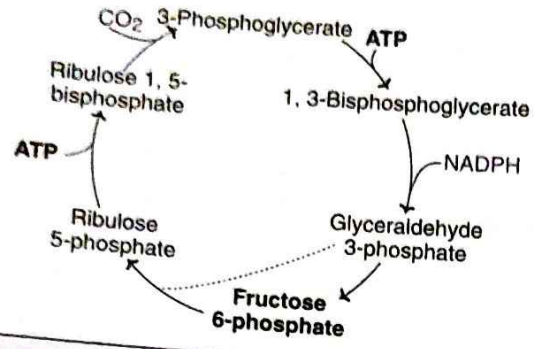


Fig. 67.8 : An outline of the Calvin cycle of photosynthesis.

chloroplasts. This results in the production of excitation energy which is transferred from one chlorophyll molecule to another, until it is trapped by a reaction center. The light-activated transfer of an electron to an acceptor (photosystems) occurs at the reaction center.

Photosynthesis primarily requires the interactions of two distinct photosystems (I and II). Photosystem I generates a strong reductant that results in the formation of NADPH. Photosystem II produces a strong oxidant that forms O_2 from H_2O. Further, the generation of ATP occurs as electrons flow from photosystem II to photosystem I (***Fig. 67.8***). Thus, light is responsible for the flow of electrons from H_2O to NADPH with a concomitant generation of ATP.

The Calvin cycle

The ***dark phase of photosynthesis*** is referred to as Calvin cycle. In this cycle, the ATP and NADPH produced in the light reaction (described above) are utilized to convert CO_2 to hexoses and other organic compounds (***Fig. 67.9***). The Calvin cycle starts with a reaction of CO_2 and ribulose 1, 5-bisphosphate to form two molecules 3-phosphoglycerate. This 3-phosphoglycerate can be converted to fructose 6-phosphate, glucose 6-phosphate and other carbon compounds.

METABOLISM OF LIPIDS

Lipids are indispensable for cell structure and function. Due to their hydrophobic and non-polar nature, lipids differ from rest of the body compounds and are unique in their action.

The important pathways/cycles of lipid metabolism are briefly described.

FATTY ACID OXIDATION

The fatty acids in the body are mostly oxidized by ***β-oxidation***. β-Oxidation may be defined as the ***oxidation of fatty acids on the β-carbon atom***. This results in the sequential removal of a two carbon fragment, acetyl CoA (***Fig. 67.10***).

Fatty acid oxidation— stages and tissues

The β-oxidation of fatty acids involves three stages

I. Activation of fatty acids occurring in the cytosol;

II. Transport of fatty acids into mitochondria;

III. β-Oxidation proper in the mitochondrial matrix.

Fatty acids are oxidized by most of the tissues in the body. However, ***brain, erythrocytes and adrenal medulla cannot utilize fatty acids for energy requirement.***

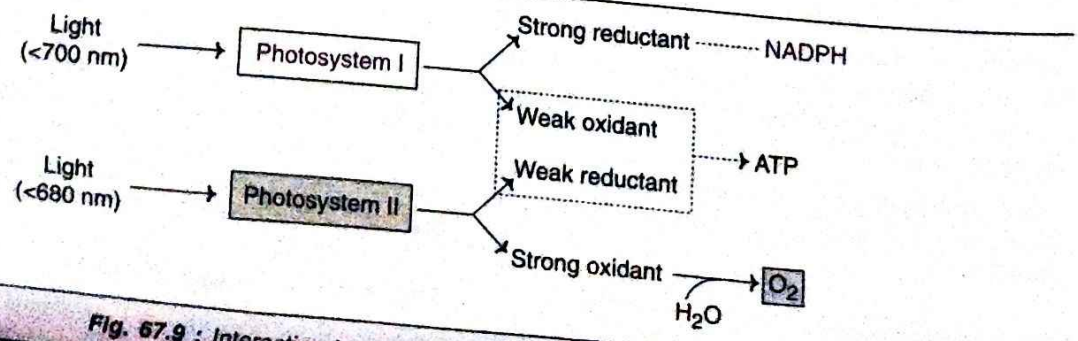


Fig. 67.9 : Interaction between two photosystems in photosynthesis.

BIOSYNTHESIS OF FATTY ACIDS

The dietary carbohydrates and amino acids, when consumed in excess, can be converted to fatty acids and stored as triacylglycerols. ***De novo*** (new) synthesis of fatty acids occurs predominantly in liver, kidney, adipose tissue and lactating mammary glands. The ***enzyme machinery*** for fatty acid production is located in the ***cytosomal fraction*** of the cell. Acetyl CoA is the source of carbon atoms while NADPH provides the reducing equivalents and ATP supplies energy for fatty acid formation. The fatty acid synthesis may be learnt in 2 stages

I. Conversion of acetyl CoA to malonyl CoA.

II. Reactions of fatty acid synthase complex.

I. Formation of malonyl CoA

Acetyl CoA is carboxylated to malonyl CoA by the enzyme ***acetyl CoA carboxylase.*** This is an ATP-dependent reaction and requires ***biotin*** for CO_2 fixation. Acetyl CoA carboxylase is a regulatory enzyme in fatty acid synthesis.

II. Reactions of fatty acid synthase complex

The remaining reactions of fatty acid synthesis are catalysed by a multifunctional enzyme known as ***fatty acid synthase (FAS) complex.*** In eukaryotic cells, including man, the fatty acid synthase exists as a dimer with two identical units. Each monomer possesses the activities of seven different enzymes and an ***acyl carrier protein (ACP)*** bound to 4'-phosphopantetheine. Fatty acid synthase functions as a single unit catalysing all the seven reactions. Dissociation of the synthase complex results in loss of the enzyme activities.

The sequence of reactions for the extra—mitochondrial synthesis of fatty acids (palmitate) is depicted in ***Fig. 67.11***, and described below

1. The two carbon fragment of acetyl CoA is transferred to ACP of fatty acid synthase, catalysed by the enzyme, ***acetyl CoA-ACP transacylase***. The acetyl unit is then transferred from ACP to cysteine residue of the enzyme. Thus ACP site falls vacant.

2. The enzyme ***malonyl CoA-ACP transacylase*** transfers malonate from malonyl CoA to bind to ACP.

3. The acetyl unit attached to cysteine is transferred to malonyl group (bound to ACP). The malonyl moiety loses CO_2 which was added by ***acetyl CoA carboxylase.*** Thus, CO_2 is never incorporated into fatty acid carbon chain.

4. β-Ketoacyl ACP reductase reduces ketoacyl group to hydroxyacyl group. The reducing equivalents are supplied by NADPH (from HMP shunt).

5. β-Hydroxyacyl ACP undergoes dehydration. A molecule of water is eliminated and a double bond is introduced between α and β carbons.

6. A second NADPH-dependent reduction, catalysed by ***enoyl-ACP reductase*** occurs to produce acyl-ACP. The four-carbon unit attached to ACP is butyryl group.

The carbon chain attached to ACP is transferred to cysteine residue and the reactions 2-6 are repeated 6 more times. Each time, the fatty acid chain is lengthened by a two-carbon unit (obtained from malonyl CoA). At the end of 7 cycles, the fatty acid synthesis is complete and a 16-carbon fully saturated fatty acid—namely palmitate—bound to ACP is produced.

7. The enzyme palmitoyl thioesterase separates palmitate from fatty acid synthase. This completes the synthesis of palmitate.

METABOLISM OF CHOLESTEROL

Cholesterol is found exclusively in animals, hence it is often called as animal sterol. The total body content of cholesterol in an adult man weighing 70 kg is about 140 g i.e., around 2 g/kg body weight. Cholesterol is ***amphipathic*** in nature, since it possesses both hydrophilic and hydrophobic regions in the structure.

CHOLESTEROL BIOSYNTHESIS

About 1 g of cholesterol is synthesized per day in adults. Almost all the tissues of the body participate in cholesterol biosynthesis. The largest contribution is made by ***liver*** (50%), ***intestine*** (15%), skin, adrenal cortex, reproductive tissue etc.

The enzymes involved in cholesterol synthesis are found in the ***cytosol*** and ***microsomal fractions*** of the cell. Acetate of ***acetyl CoA provides*** all the ***carbon atoms*** in cholesterol. The reducing equivalents are supplied by ***NADPH*** while ***ATP*** provides energy. The key intermediates of cholesterol formation are depicted in ***Fig. 67.12.***

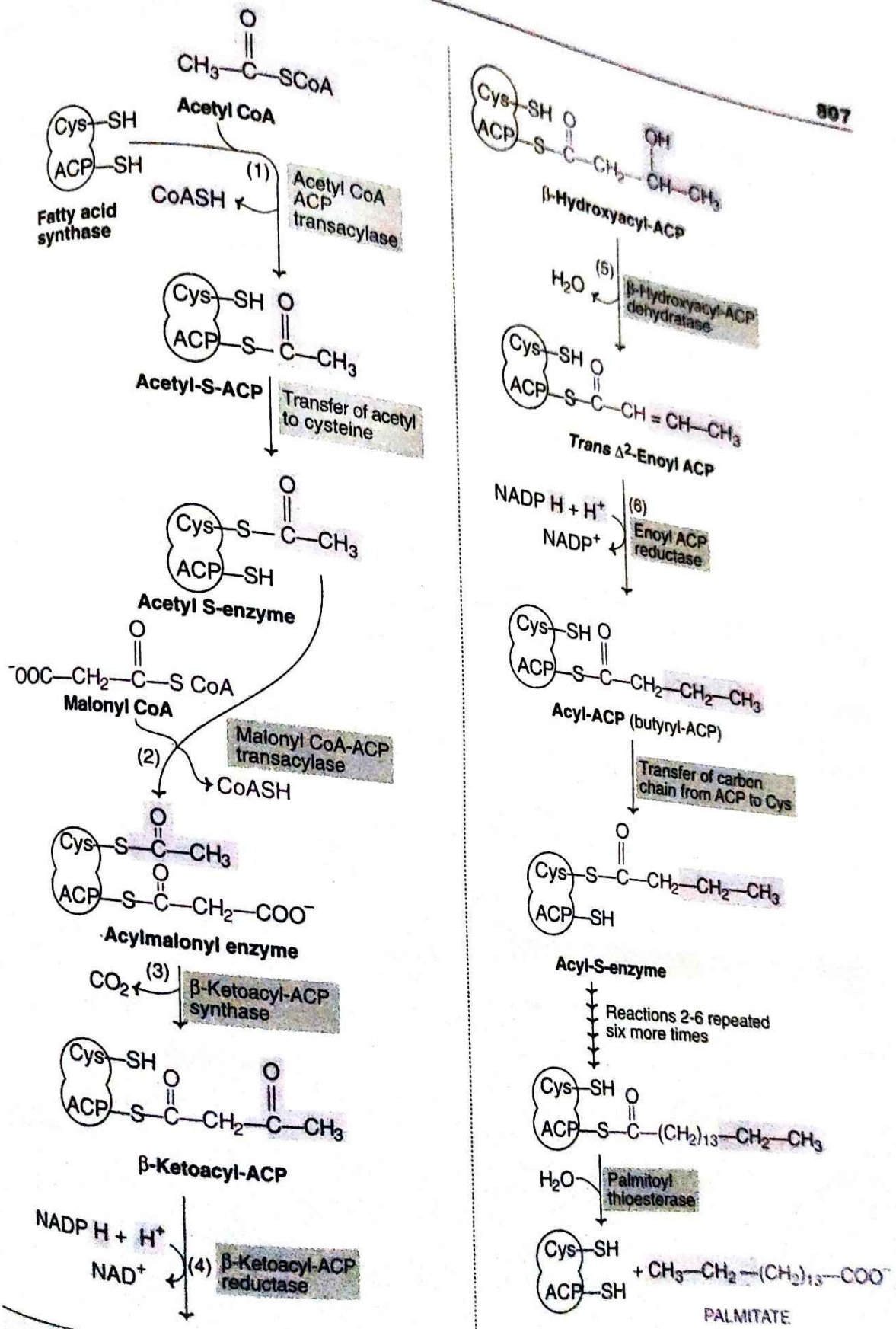


Fig. 67.11 contd. next column

Fig. 67.11 : *Biosynthesis of long chain fatty acid–palmitate. (Cys–Cysteine; ACP–Acyl carrier protein; The pathway repeats 7 times to produce palmitate; the first two carbons at the methyl end are directly from acetyl CoA, the rest of the carbons come from malonyl CoA).*

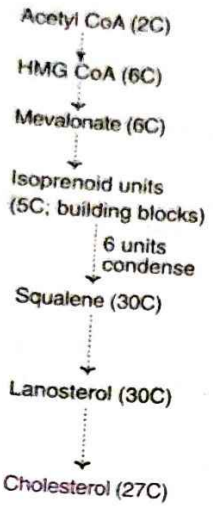


Fig. 67.12 : Outline of cholesterol biosynthesis.

DEGRADATION OF CHOLESTEROL

The steroid nucleus (ring structure) of the cholesterol cannot be degraded to CO_2 and H_2O. Cholesterol (50%) is converted to bile acids, excreted in feces, serves as a precursor for the synthesis of steroid hormones, vitamin D, coprostanol and cholestanol. The latter two are the fecal sterols, besides cholesterol.

METABOLISM OF AMINO ACIDS

Proteins are the ***most abundant organic compounds*** and constitute a major part of the body dry weight (10-12 kg in adults). They perform a wide variety of static (structural) and dynamic (enzymes, hormones, clotting factors, receptors etc.) functions. About half of the body protein (predominantly collagen) is present in the supportive tissue (skeleton and connective) while the other half is intracellular.

The proteins on degradation (proteolysis) release individual amino acids. Amino acids are not just the structural components of proteins. Each of the 20 naturally occurring amino acids undergoes its own metabolism and performs specific functions. Some of the amino acids also serve as precursors for the synthesis of many biologically important compounds (e.g. melanin, serotonin, creatine etc.).

Protein metabolism is more appropriately learnt as metabolism of amino acids.

METABOLISM OF AMINO ACIDS — GENERAL ASPECTS

The amino acids undergo certain common reactions like ***transamination*** followed by ***deamination*** for the liberation of ***ammonia***. The amino group of the amino acids is utilized for the formation of ***urea*** which is an excretory end product of protein metabolism. The carbon skeleton of the amino acids is first converted to keto acids (by transamination) which meet one or more of the following fates.

1. Utilized to generate energy.
2. Used for the synthesis of glucose.
3. Diverted for the formation of fat or ketone bodies.
4. Involved in the production of non-essential amino acids.

A general picture of amino acid metabolism is depicted in ***Fig. 67.13***.

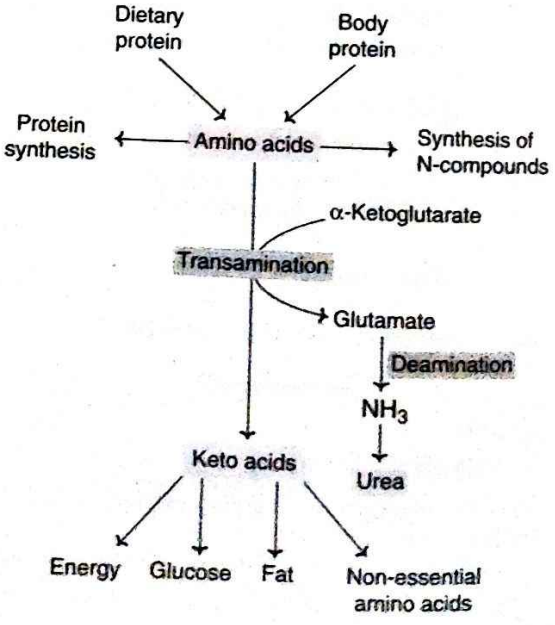


Fig. 67.13 : An overview of amino acid metabolism.

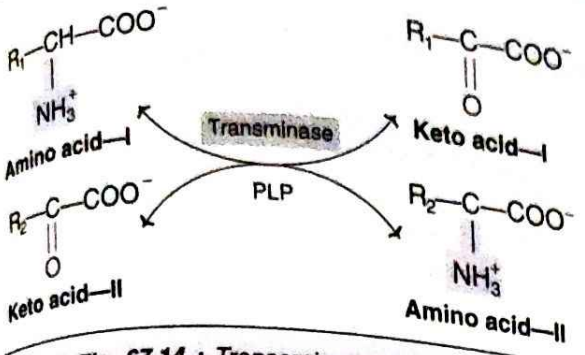


Fig. 67.14 : Transamination reaction.

TRANSAMINATION

The transfer of an amino ($-NH_2$) group from an amino acid to a keto acid is known as transamination (***Fig. 67.14***). This process involves the interconversion of a pair of amino acids and a pair of keto acids, catalysed by a group of enzymes called ***transaminases*** (recently, ***aminotransferases***).

The salient features of transamination are :

1. All transaminases require ***pyridoxal phosphate*** (PLP), a coenzyme derived from vitamin B_6.

2. There is no free NH_3 liberated, only the transfer of amino group occurs.

3. Transamination is ***reversible***.

4. It involves both catabolism (degradation) and anabolism (synthesis) of amino acids. Transamination is ultimately responsible for the synthesis of non-essential amino acids.

5. Transamination diverts the excess amino acids towards ***energy generation***.

6. The amino acids undergo transamination to finally concentrate nitrogen in glutamate. ***Glutamate*** is the only amino acid that undergoes oxidative deamination to a significant extent to liberate free NH_3 for urea synthesis.

7. All amino acids except lysine, threonine, proline and hydroxyproline participate in transamination.

DEAMINATION

The removal of amino group from the amino acids as NH_3 is deamination. It results in the liberation of ammonia for urea synthesis. Deamination may be either oxidative or non-oxidative.

UREA CYCLE

Urea is the end product of protein metabolism (amino acid metabolism). The nitrogen of amino acids converted to ammonia (as described above) is toxic to the body. It is converted to urea and detoxified. As such, urea accounts for 80-90% of the nitrogen containing substances excreted in urine.

Urea is ***synthesized in liver*** and transported to kidneys for excretion in urine. Urea cycle is the ***first metabolic cycle*** that was elucidated by Hans Krebs and Kurt Henseleit (1932), hence it is known as ***Krebs-Henseleit cycle***. The individual reactions, however, were described in more detail later on by Ratner and Cohen.

Urea has ***two amino ($-NH_2$) groups***, one derived ***from NH_3*** and the other ***from aspartate***. Carbon atom is supplied by CO_2. Urea synthesis is a five-step cyclic process, with five distinct enzymes. The first ***two enzymes*** are present in ***mitochondria*** while the ***rest*** are localized in ***cytosol***. The reactions of urea cycle are depicted in ***Fig. 67.15***.

METABOLISM OF INDIVIDUAL AMINO ACIDS

The metabolisms of certain individual amino acids are very briefly given in the form of overviews.

GLYCINE

Glycine (Gly, G) is a non-essential, optically inactive and ***glycogenic*** (precursor for glucose) amino acid. It is indispensable for chicks. The outline of glycine metabolism is depicted in ***Fig. 67.16***. Glycine is actively involved in the synthesis of many specialized products (heme, purines, creatine etc.) in the body, besides its incorporation into proteins, synthesis of serine and glucose and participation in one-carbon metabolism.

PHENYLALANINE AND TYROSINE

Phenylalanine (Phe, F) and tyrosine (Tyr, Y) are structurally related aromatic amino acids. Phenylalanine is an essential amino acid while tyrosine is non-essential. Besides its incorporation into proteins, the only function of phenylalanine is its conversion to tyrosine. For this reason, ingestion of tyrosine can reduce the dietary requirement of phenylalanine. This phenomenon is referred to as ***'sparing action' of tyrosine on phenylalanine.***

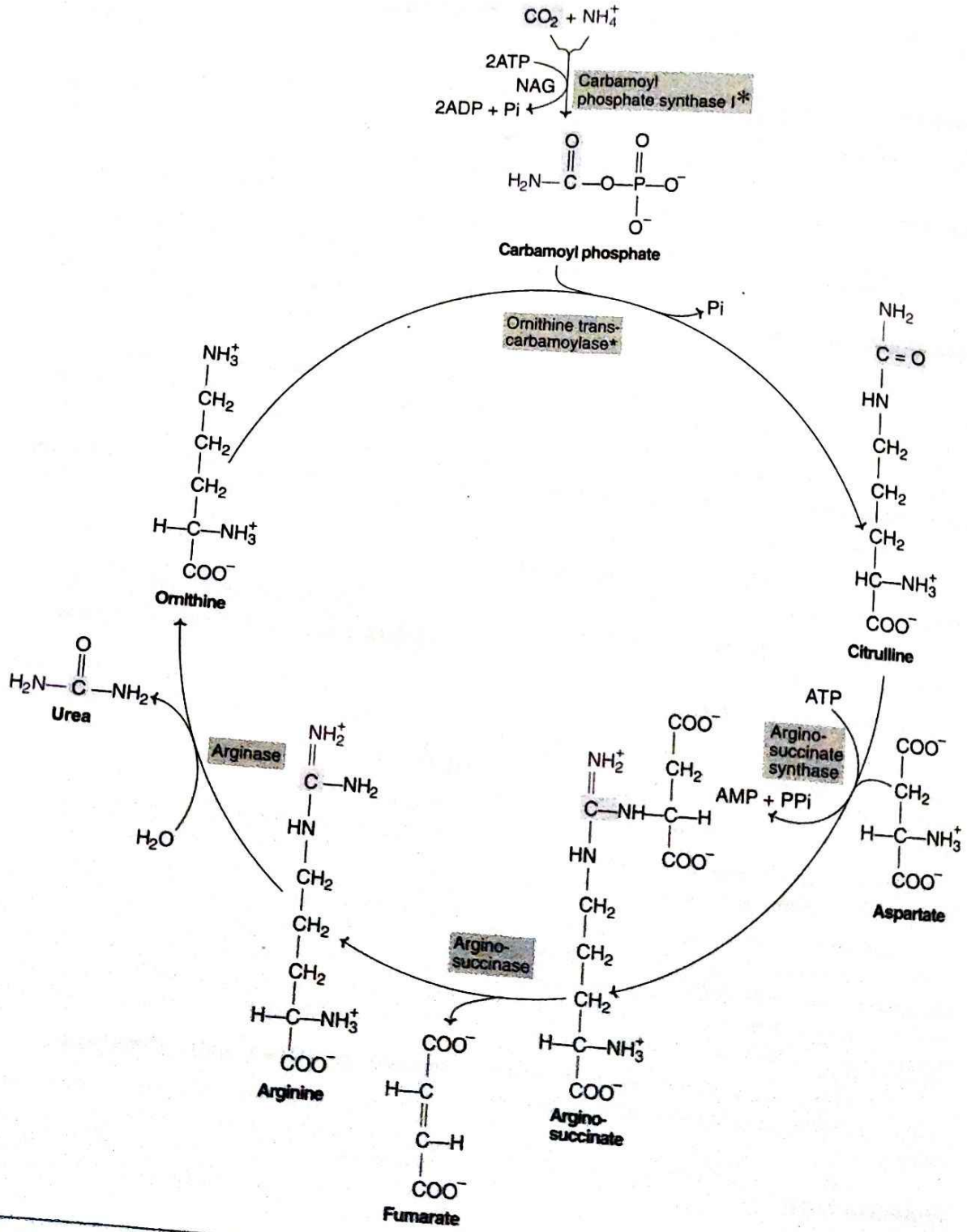


Fig. 67.15 : *Reactions of urea cycle (NAG–N-acetylglutamate; (in the formation of urea, one amino group is derived from free ammonium ion while the other is from aspartate; carbon is obtained from CO_2; * mitochondrial enzymes, the rest of the enzymes are cytosomal).*

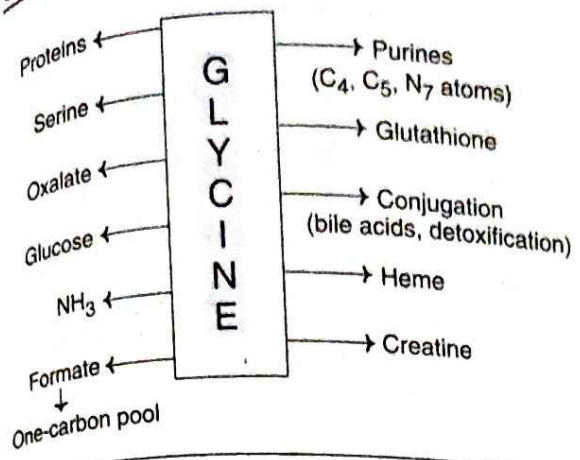


Fig. 67.16 : Overview of glycine metabolism.

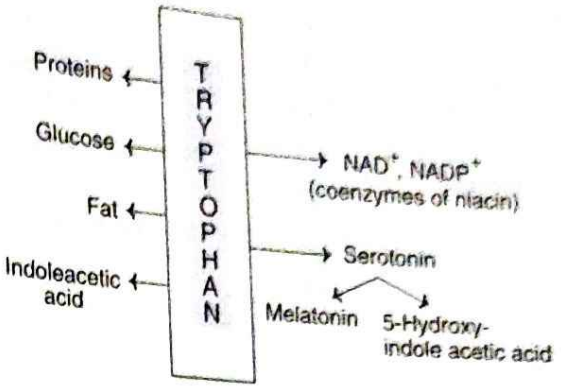


Fig. 67.18 : Overview of tryptophan metabolism.

The predominant metabolism of phenylalanine occurs through tyrosine. Tyrosine is incorporated into proteins and is involved in the synthesis of a variety of biologically important compounds—***epinephrine, norepinephrine, dopamine*** (catecholamines), ***thyroid hormones***—and the pigment ***melanin*** (***Fig. 67.17***). During the course of degradation, phenylalanine and tyrosine are converted to metabolites which can serve as precursors for the synthesis of glucose and fat. Hence, these amino acids are both ***glucogenic*** and ***ketogenic***.

TRYPTOPHAN

Tryptophan (Trp, W) was the first to be identified as an essential amino acid. It contains an indole ring and chemically it is α-amino β-indole propionic acid. Tryptophan is both ***glucogenic*** and ***ketogenic*** in nature. It is a precusor for the synthesis of important compounds, namely ***NAD^+*** and ***$NADP^+$*** (coenzymes of niacin), ***serotonin*** and ***melatonin*** (***Fig. 67.18***).

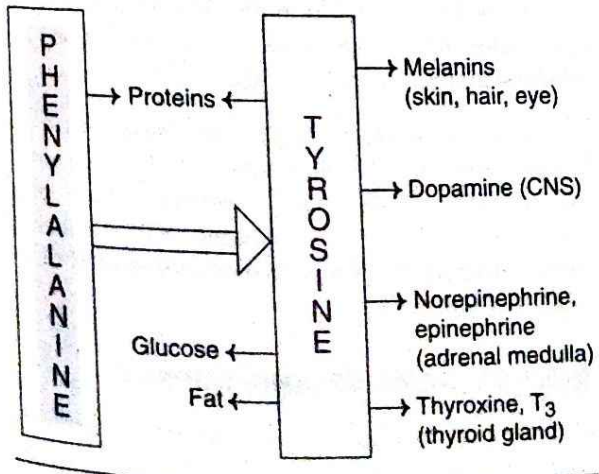


Fig. 67.17 : Overview of phenylalanine and tyrosine metabolism (CNS—Central nervous system; T_3—Triiodothyronine).

SULFUR AMINO ACIDS

The sulfur-containing amino acids are methionine, cysteine and cystine. Among these, only ***methionine*** is ***essential***. It serves as a precursor for the synthesis of cysteine and cystine which are, therefore, non-essential. An overview of the metabolism of the sulfur amino acids is depicted in ***Fig. 67.19***.

GLUTAMATE AND GLUTAMINE

Glutamate and glutamine are non-essential glycogenic amino acids. Both of them play a predominant role in the amino acid metabolism and are directly involved in the final transfer of amino group for urea synthesis. In ***Fig. 67.20***, an outline of glutamate and glutamine metabolism is given.

FATE OF CARBON SKELETON OF AMINO ACIDS

After the removal of amino groups, the carbon skeleton of amino acids is converted to intermediates of TCA cycle or their precursors. The carbon skeleton finally has one or more of the following fates :

1. Oxidation via TCA cycle to produce energy (about 10-15% of body needs).

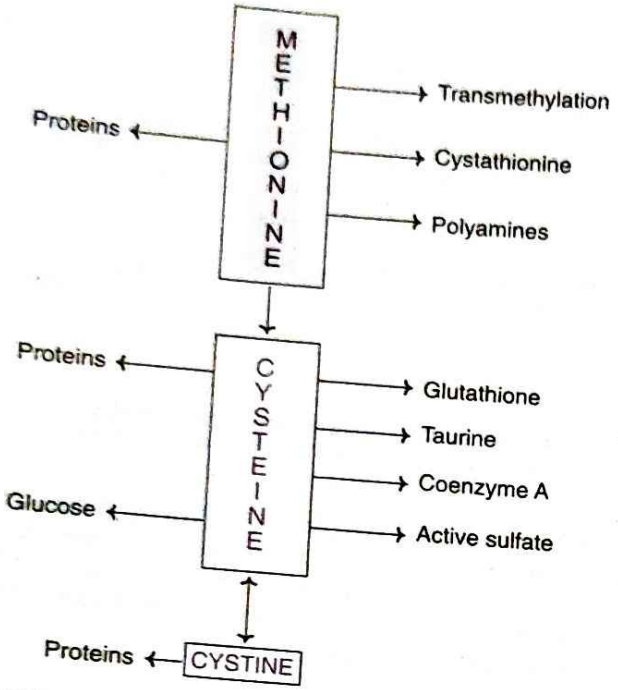


Fig. 67.19 : Overview of the metabolism of sulfur amino acids.

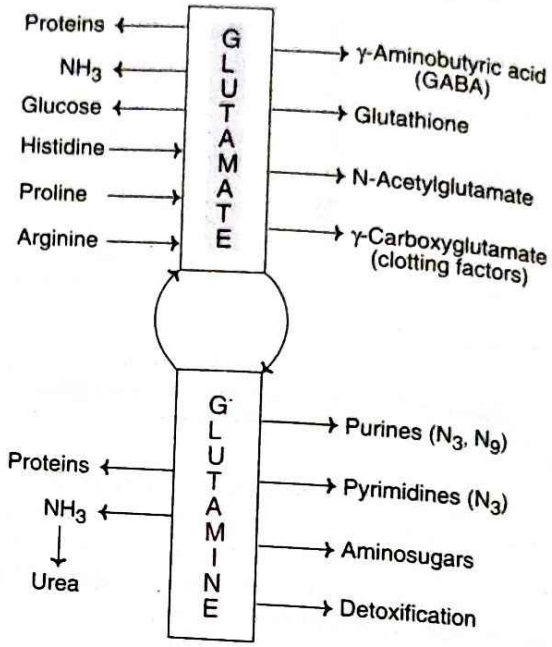


Fig. 67.20 : Overview of glutamate and glutamine metabolism.

2. Synthesis of glucose.

3. Formation of lipids—fatty acids and ketone bodies.

4. Synthesis of non-essential amino acids.

The carbon skeletons of the 20 standard amino acids (or the amino acids of proteins) are degraded to one of the following seven products—***pyruvate***, ***α-ketoglutarate***, ***succinyl CoA***, ***fumarate***, ***oxaloacetate***, ***acetyl CoA*** and ***acetoacetate***. Some authors use the term amphibolic (*Greek*: amphiboles—uncertain) intermediates to these compounds due to their multiple metabolic functions.

The amino acids are classified into three groups, based on the nature of the metabolic end products of carbon skeleton (***Table 67.1***).

Inborn errors of amino acid metabolism—a summary

Several inherited disorders are associated with amino acid metabolism. In ***Table 67.2***, a summary of major diseases and the enzyme defects is given.

INTEGRATION OF METABOLISM

Metabolism is a continuous process, with thousands of reactions, simultaneously occurring in the living cell. However, biochemists prefer to present metabolism in the form of reactions and metabolic pathways. This is done for the sake of convenience in presentation and understanding.

In the preceeding pages, we have learnt the metabolism of carbohydrates, lipids and amino acids. We shall now consider the organism as a whole and integrate the metabolism with particular reference to energy demands of the body organism.

ENERGY DEMAND AND SUPPLY

The organisms possess variable energy demands, hence the supply (input) is also equally variable. The consumed metabolic fuel may be burnt (oxidized to CO_2 and H_2O) or stored to meet the energy requirements as per the body needs. ***ATP***

TABLE 67.1 Classification of amino acids based on the fate of carbon skeleton

Glycogenic (glucogenic)	*Glycogenic and Ketogenic*	*Ketogenic*
Alanine	Phenylalanine*	Leucine*
Arginine*	Isoleucine*	Lysine*
Aspartate	Tyrosine	
Cysteine	Tryptophan*	
Glutamine		
Glutamate		
Glycine		
Histidine*		
Hydroxyproline		
Methionine*		
Proline		
Serine		
Threonine*		
Valine*		

**Essential amino acids; (Helpful tips to recall—ketogenic amino acids start with letter 'L'; PITT for glyco-and ketogenic amino acids; rest of the 20 amino acids are only glycogenic).*

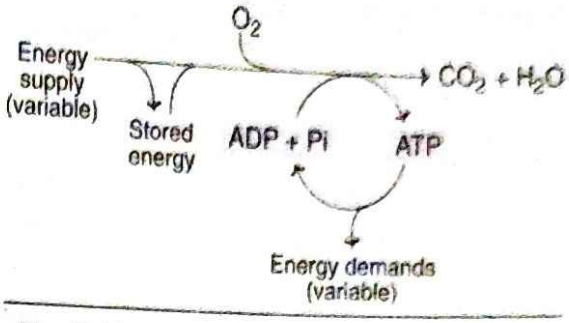


***Fig. 67.21** : A summary of body's energy supply and demands. (**Note** : ATP serves as the energy currency).*

serves as the ***energy currency of the cell*** in this process (***Fig. 67.21***).

The humans possess enormous capacity for food consumption. It is estimated that one can consume as much as 100 times his/her basal requirements! Obesity, a disorder of overnutrition mostly prevalent in affluent societies, is primarily a consequence of overconsumption.

INTEGRATION OF MAJOR METABOLIC PATHWAYS OF ENERGY METABOLISM

An overview of the interrelationship between the important metabolic pathways, concerned with fuel metabolism depicted in ***Fig. 67.22***, is briefly

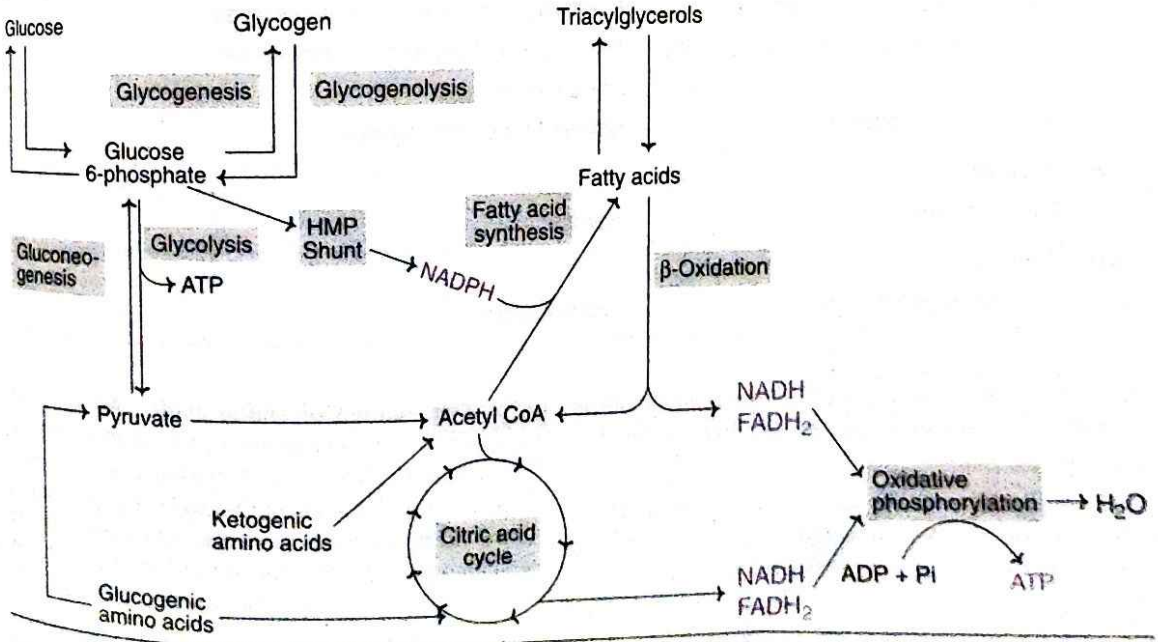


***Fig. 67.22** : An overview of integration of metabolic pathways of energy metabolism (HMP shunt—Hexose monophosphate shunt).*

Table 67.2 Inborn errors of amino acid metabolism

Disorder	*Metabolic defect (enzyme/other)*
I. Phenylalanine and tyrosine	
1. Phenylketonuria	Phenylalanine hydroxylase
2. Tyrosinemia type II	Tyrosine transaminase
3. Neonatal tyrosinemia	p-Hydroxy phenylpyruvate dioxygenase
4. Alkaptonuria	Homogentisate oxidase
5. Tyrosinosis (tyrosinemia type I)	Maleyl acetoacetate isomerase or fumaryl acetoacetate hydrolase
6. Albinism	Tyrosinase
II. Sulfur amino acids (methionine, cysteine and cystine)	
7. Cystinuria	Defect in renal reabsorption
8. Cystinosis	Impairment in cystine utilization (defect in lysosomal function)
9. Homocystinuria type I	Cystathionine synthetase
10. Homocystinuria type II	N^5, N^{10}-Methylene THF reductase
11. Homocystinuria type III	N^5-Methyl THF-homocysteine methyltransferase
12. Cystathionuria	Cystathioninase
III. Glycine	
13. Glycinuria	Defect in renal reabsorption
14. Primary hyperoxaluria	Glycine transaminase
IV. Tryptophan	
15. Hartnup's disease	Defective intestinal absorption
V. Branched chain amino acids (valine, leucine and isoleucine)	
16. Maple syrup urine disease	Branched chain α-keto acid dehydrogenase
17. Intermittent branched chain ketonuria	Variant of the above enzyme (less severe)
18. Hypervalinemia	Valine transaminase
19. Isovaleric acidemia	Isovaleryl CoA dehydrogenase
VI. Histidine	
20. Histidinemia	Histidase
VII. Proline	
21. Hyperprolinemia type I	Proline oxidase

described here. For detailed information on these metabolic pathways, the reader must refer the preceeding pages.

1. **Glycolysis :** The degradation of glucose to pyruvate (lactate under anaerobic condition) generates 8 ATP. Pyruvate is converted to acetyl CoA.

2. **Fatty acid oxidation :** Fatty acids undergo sequential degradation with a release of 2-carbon fragment, namely acetyl CoA. The energy is trapped in the form of NADH and $FADH_2$.

3. **Degradation of amino acids :** Amino acids, particularly when consumed in excess than required for protein synthesis, are degraded and utilized to meet the fuel demands of the body. The glucogenic amino acids can serve as precursors for the synthesis of glucose via the formation of pyruvate or intermediates of citric acid cycle. The ketogenic amino acids are the precursors for acetyl CoA.

4. **Citric acid cycle :** Acetyl CoA is the key and common metabolite, produced from different fuel

sources (carbohydrates, lipids, amino acids). ***Acetyl CoA*** enters citric acid cycle and gets oxidized to CO_2. Thus, citric acid cycle is the final common metabolic pathway for the oxidation of all foodstuffs. Most of the energy is trapped in the form of NADH and $FADH_2$.

5. **Oxidative phosphorylation :** The NADH and $FADH_2$, produced in different metabolic pathways, are finally oxidized in the electron transport chain (ETC). The ETC is coupled with oxidative phosphorylation ***to generate ATP.***

6. **Hexose monophosphate shunt :** This pathway is primarily concerned with the liberation of NADPH and ribose sugar. NADPH is utilized for the biosynthesis of several compounds, including fatty acids. Ribose is an essential component of nucleotides and nucleic acids (note—DNA contains deoxyribose).

7. **Gluconeogenesis :** The synthesis of glucose from non-carbohydrate sources constitutes gluconeogenesis. Several compounds (e.g. pyruvate, glycerol, amino acids) can serve as precursors for gluconeogenesis.

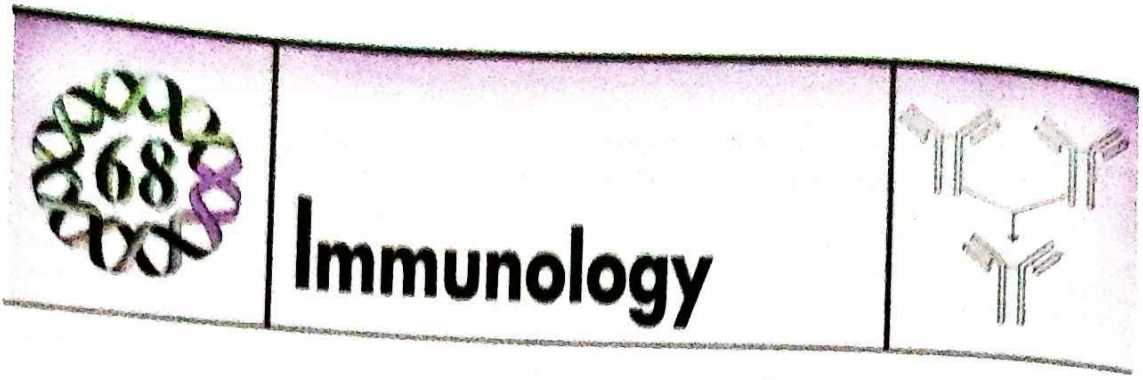
8. **Glycogen metabolism :** Glycogen is the storage form of glucose, mostly found in liver and muscle. It is degraded (glycogenolysis) and synthesized (glycogenesis) by independent pathways. Glycogen effectively serves as a ***fuel reserve*** to meet body needs, ***for a brief period*** (between meals).

REGULATION OF METABOLIC PATHWAYS

The metabolic pathways, in general, are controlled by four different mechanisms

1. The availability of substrates
2. Covalent modification of enzymes
3. Allosteric regulation
4. Regulation of enzyme synthesis.

The details of these regulatory processes are discussed under the individual metabolic pathways, in the preceeding pages.



68 Immunology

Immunology deals with the study of immunity and immune systems of vertebrates. Immunity (immunis literally means exempt/free from burden) broadly involves the ***resistance shown***, and ***protection offered by the host organism against the infectious diseases***. The immune system consists of a complex network of cells and molecules, and their interactions. It is specifically designed to eliminate infectious organisms from the body. This is possible since the organism is capable of distinguishing the self from non-self, and eliminate non-self.

Immunity is broadly divided into two types — innate (non-specific) immunity and adaptive or acquired (specific) immunity.

INNATE IMMUNITY

Innate immunity is non-specific, and represents the inherent capability of the organism to offer resistance against diseases. It consists of defensive barriers.

First line of defense

The ***skin*** is the largest organ in the human body, constituting about 15% of the adult body weight. The skin provides mechanical barrier to prevent the entry of microorganisms and viruses. The acidic (pH 3-5) environment on the skin surface inhibits the growth of certain microorganisms. Further, the sweat contains an enzyme lysozyme that can destroy bacterial cell wall.

Second line of defense

Despite the physical barriers, the micro-organisms do enter the body. The body defends itself and ***eliminates the invading organisms by non-specific mechanisms*** such as sneezing and secretions of the mucus. In addition, the body also tries to kill the pathogens by phagocytosis (involving macrophages and complement system). The inflammatory response and fever response of the body also form a part of innate immunity.

THE IMMUNE SYSTEM

The immune system represents the third and most potent defense mechanism of the body. Acquired (adaptive or specific) immunity is capable of specifically recognizing and eliminating the invading microorganisms and foreign molecules (antigens). In contrast to innate immunity, the acquired immunity displays four distinct characteristics

- Antigen specificity
- Recognition diversity
- Immunological memory
- Discrimination between self and non-self.

The ***body*** possess tremendous capability to specifically ***identify various antigens*** (antigen is a foreign substance, usually a protein or a

carbohydrate that elicits immune response). Exposure to an antigen leads to the development of immunological memory. As a result, a second encounter of the body to the same antigen results in a heightened state of immune response. The immune system recognizes and responds to foreign antigens as it is capable of distinguishing self and non-self. Autoimmune diseases are caused due to a failure to discriminate self and non-self antigens.

ORGANIZATION OF IMMUNE SYSTEM

The immune system consists of several organs distributed throughout the body (***Fig. 68.1***). These lymphoid organs are categorized as primary and secondary.

Primary lymphoid organs

These organs provide appropriate micro-environment for the development and maturation of antigen-sensitive lymphocytes (a type of white blood cells). The ***thymus*** (situated above the heart) and bone marrow are the central or primary lymphoid organs. T-lymphocyte maturation occurs in the thymus while B-lymphocyte maturation takes place in the bone marrow.

Secondary lymphoid organs

These are the sites for the initiation of immune response. e.g. spleen, tonsils, lymph nodes, appendix, Peyers patches in the gut. Secondary lymphoid organs provide the microenvironment for interaction between antigens and mature lymphocytes.

CELLS OF THE IMMUNE SYSTEM

Two types of lymphocytes namely B-cells and T-cells are critical for the immune system. In addition, several accessory cells and effector cells also participate.

B-lymphocytes

The site of development and maturation of B-cells occurs in ***b***ursa fabricus in birds, and ***b***one marrow in mammals. During the course of immune response. B-cells mature into plasma cells and secrete antibodies (immunoglobulins).

The B-cells possess the capability to specifically recognize each antigen and produce antibodies against it. B-lymphocytes are inimately associated with **humoral immunity**.

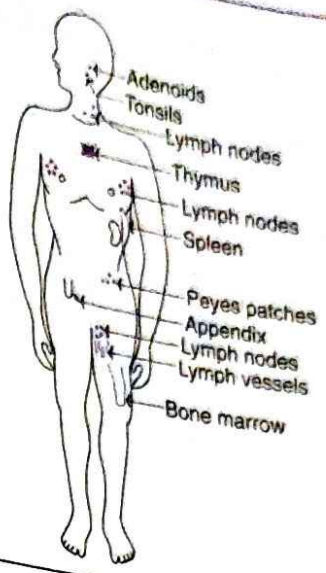


Fig. 68.1 : A diagrammatic representation of human lymphatic system.

T-lymphocytes

The maturation of T-cells occurs in the thymus, hence the name. The T-cells can identify viruses and microorganisms from the antigens displayed on their surfaces. There are at least four different types of T-cells.

- ***Inducer T-cells*** that mediate the development of T-cells in the thymus.
- ***Cytotoxic T-cells*** (T_C), capable of recognizing and killing the infected or abnormal cells.
- ***Helper T-cells*** (T_H) that initiate immune responses.
- ***Suppressor T-cells*** mediate the suppression of immune response.

T-lymphocytes are responsible for the ***cell-mediated immunity***.

IMMUNOGLOBULINS

The humoral immunity is mediated by a special group of proteins called immunoglobulins or ***antibodies***, produced by B-lymphocytes.

STRUCTURE OF IMMUNOGLOBULINS

All the immunoglobulin (Ig) molecules basically consist of two identical ***heavy (H) chains*** (mol. wt. 53,000 to 75,000 each) and two identical ***light (L) chains*** (mol. wt. 23,000 each) held together by disulfide linkages and non-covalent interactions.

Table 68.1 Characteristics of human immunoglobulins

Type	H-Chain	L-Chains	Molecular formula	Molecular weight	Percentage carbohydrate	Serum conc. mg/dl	Major function(s)
IgG	γ	κ or λ	$\gamma_2\kappa_2$ or $\gamma_2\lambda_2$	~150,000	3	800–1,500	Mostly responsible for humoral immunity
IgA	α	κ or λ	$(\alpha_2\kappa_2)_{1-3}$ or $(\alpha_2\lambda_2)_{1-3}$	~$(160,000)_{1-3}$	8	150–400	Protects the body surfaces
IgM	μ	κ or λ	$(\mu_2\kappa_2)_5$ or $(\mu_2\lambda_2)_5$	~900,000	12	50–200	Humoral immunity, serves as first line of defense
IgD	δ	κ or λ	$(\delta_2\kappa_2$ or $\delta_2\lambda_2)$	~180,000	13	1–10	B-cell receptor?
IgE	ε	κ or λ	$\varepsilon_2\kappa_2$ or $\varepsilon_2\lambda_2$	~190,000	12	0.02–0.05	Humoral sensitivity and histamine release.

(***Fig. 68.2***). Thus, immunoglobulin is a Y-shaped tetramer (H_2L_2). Each heavy chain contains approximately 450 amino acids while each light chain has 212 amino acids. The heavy chains of Ig are linked to carbohydrates, hence immunoglobulins are ***glycoproteins***.

Constant and variable regions : Each chain (L or H) of Ig has two regions (domains), namely the constant and the variable. The amino terminal half of the light chain is the variable region (V_L) while the carboxy terminal half is the constant region (C_L). As regards heavy chain, approximately one-quarter of the amino terminal region is variable (V_H) while the remaining three-quarters is constant (C_{H_1}, C_{H_2}, C_{H_3}). The amino acid sequence (with its tertiary structure) of variable regions of light and heavy chains is responsible for the specific binding of immunoglobulin (antibody) with antigen.

CLASSES OF IMMUNOGLOBULINS

Humans have five classes of immunoglobulins — namely ***IgG, IgA, IgM, IgD and IgE*** — containing the heavy chains γ, α, μ, δ and ε, respectively. The type of heavy chain ultimately determines the class and the function of a given Ig.

Two types of light chains — namely kappa (κ) and lambda (λ) — are found in immunoglobulins. They differ in their structure in C_L regions. An immunoglobulin (of any class) contains two κ or two λ light chains and never a mixture. The occurrence of κ chains is more common in human immunoglobulins than λ chains.

The characteristics of the 5 classes of human immunoglobulins are given in ***Table 68.1***.

Immunoglobulin G (IgG)

IgG is the most abundant (75–80%) class of immunoglobulins. IgG is composed of a single Y-shaped unit (monomer). It can traverse blood vessels readily. IgG is the only immunoglobulin that can cross the placenta and transfer the mother's immunity to the developing fetus. IgG triggers ***foreign cell destruction*** mediated by complement system.

Immunoglobulin A (IgA)

IgA occurs as a single (monomer) or double unit (dimer) held together by J chain. It is mostly found in the body secretions such as saliva, tears, sweat, milk and the walls of intestine. IgA is the most predominant antibody in the colostrum, the initial secretion from the mother's breast after a baby is born. The IgA molecules bind with bacterial antigens present on the body (outer epithelial) surfaces and remove them. In this way, IgA ***prevents the foreign substances*** from entering the body cells.

Immunoglobulin M (IgM)

IgM is the largest immunoglobulin composed of 5 Y-shaped units (IgG type) held together by a J polypeptide chain. Thus IgM is a pentamer. Due to its large size, IgM cannot traverse blood vessels, hence it is restricted to the blood stream. IgM is the first antibody to be produced in response to an

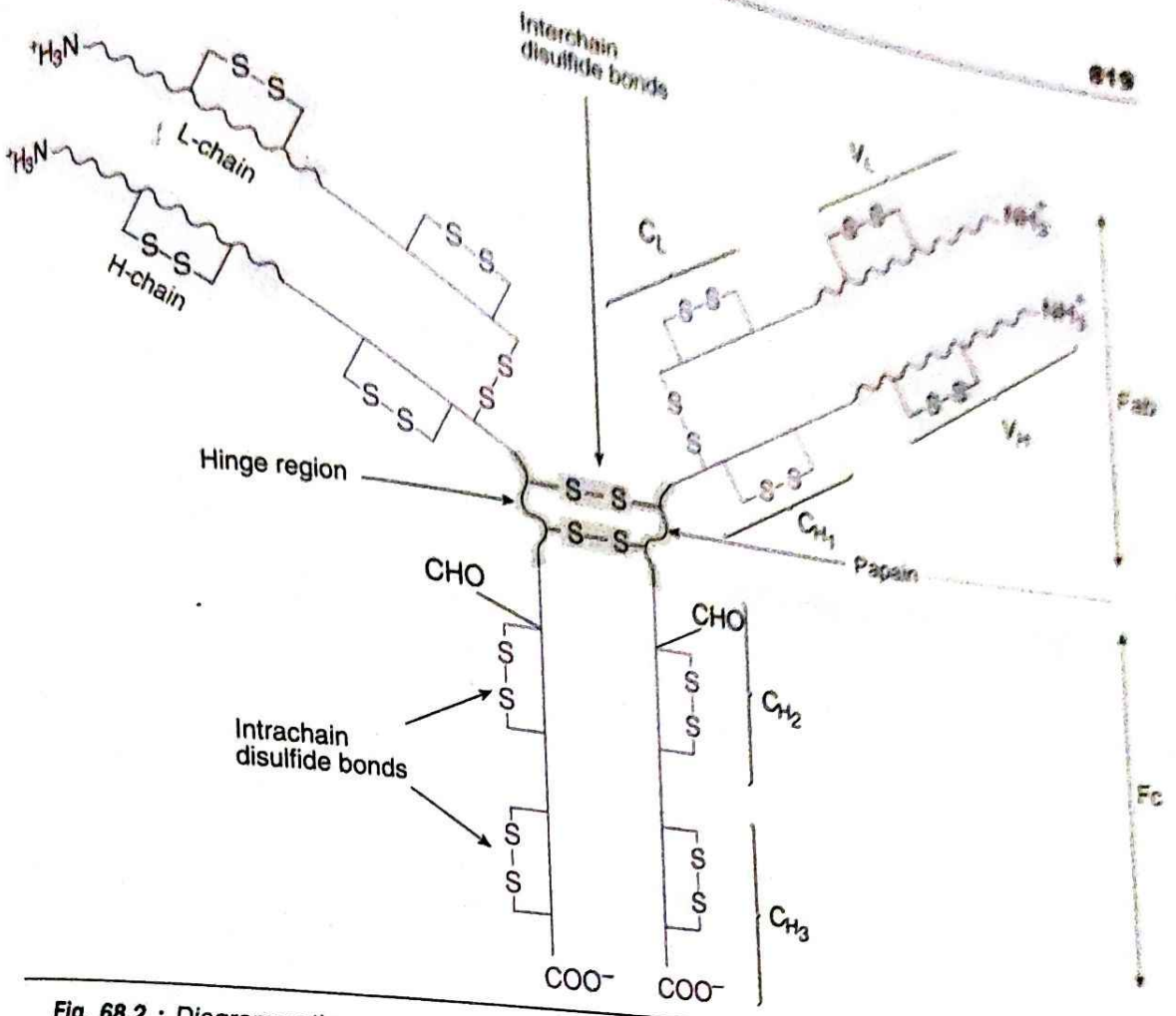


Fig. 68.2 : Diagrammatic representation of human IgG molecule (V—Variable region, C—Constant region; CHO—Carbohydrate; Each heavy chain is composed of four units—V_H, C_{H_1}, C_{H_2}, C_{H_3} while light chain consists of two units—V_L, C_L).

antigen and is the most ***effective against invading microorganisms.***

Immunoglobulin D (IgD)

IgD is composed of a single Y-shaped unit and is present in a low concentration in the circulation. IgD molecules are present on the surface of B cells. Their function, however, is not known for certain. Some workers believe that IgD may function as ***B-cell receptor.***

Immunoglobulin E (IgE)

IgE is a single Y-shaped monomer. It is normally present in minute concentration in blood. IgE levels are elevated in individuals with ***allergies*** as it is associated with the body's allergic responses. The IgE molecules tightly bind with mast cells which release histamine and cause allergy.

SYNTHESIS OF IMMUNOGLOBULINS

There are millions of different antigens. It was a big puzzle for a long time how an individual can produce so many antibodies to protect against antigens. It is now recognized that a gene rearrangement involving a combination of several genes is responsible for generating an extremely large number of antibodies. (For more details Refer Chapter 5).

MAJOR HISTOCOMPATIBILITY COMPLEX

The major histocompatibility complex (***MHC***) represents a special group of ***proteins, present on the cell surfaces of T-lymphocytes.*** MHC is involved in the recognition of antigens on T-cells. It may be noted here that the B-cell receptors (antibodies) can recognize antigens and on their own, while T-cells can do so through the mediation of MHC.

In humans, the MHC proteins are encoded by a cluster of genes located on chromosome 6 (it is on chromosome 17 for mice). The major histocompatibility complex in humans is referred to as ***human leukocyte antigen (HLA)***. Three classes of MHC molecules (chemically glycoproteins) are known in human. Class I molecules are found on almost all the nucleated cells of the body. Class II molecules are associated only with leukocytes involved in cell-mediated immune response. Class III molecules are the secreted proteins possessing immune functions e.g. complement components (C_2, C_4), tumor necrosis factor.

THE IMMUNE RESPONSE

The immune response refers to the series of reactions carried out by the immune system in the body against the foreign invader. When an infection takes place or when an antigen enters the body, it is trapped by the macrophages in lymphoid organs. The phagocytic cells which are guarding the body by constant patrolling engulf and digest the foreign substance. However, the partially digested antigen (i.e. processed antigen) with antigenic epitopes attaches to lymphocytes.

T-helper cells (T_H) play a key role the immune response (***Fig. 68.3***). This is brought out through the participation of antigen presenting cell (APC), usually a macrophage. Receptors of T_H cell bind to class II MHC-antigen complex displayed on the surface of APC. APC secretes interleukin-I, which activates the T_H cell. This activated T_H cell actively grows and divides to produce clones of T_H cells. All the T_H cells possess receptors that are specific for the MHC-antigen complex. This facilitates triggering of immune response in an exponential manner. The T_H cells secrete interleukin-2 which promotes the prolifiration of cytotoxic T cells (T_C cells) to attack the infected cells through cell-mediated immunity. Further, interleukin-2 also activates B-cells to produce immunoglobulins which perform humoral immunity.

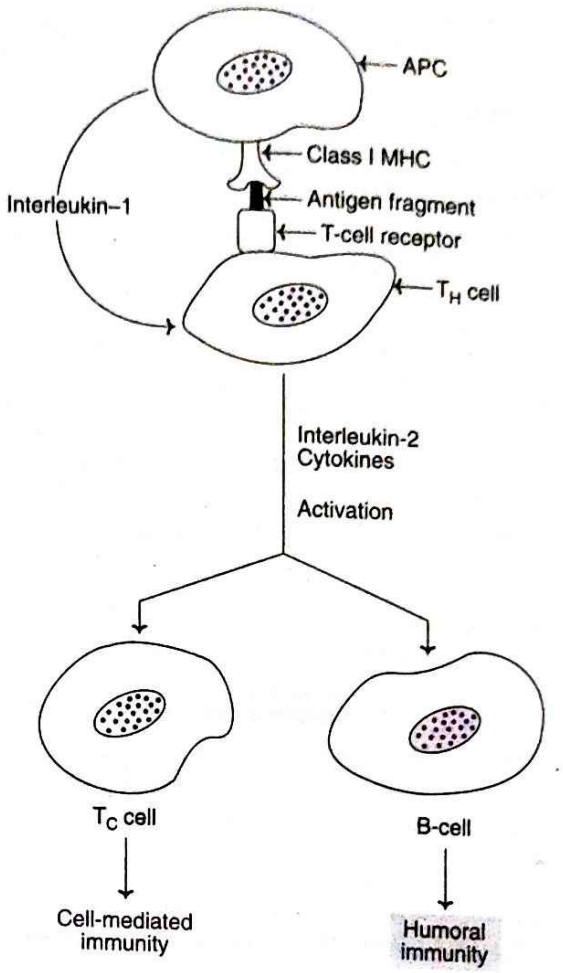


Fig. 68.3 : *A diagrammatic representation of the central role of helper T cells (T_H cells) in immune response (APC—Antigen presenting cell; T_C cell—Cytotoxic T–cell).*

CYTOKINES

Cytokines are a group of ***proteins*** that ***bring about communication between different cell types*** involved in immunity. They are low molecular weight glycoproteins and are produced by lymphoid and non-lymphoid cells during the course of immune response. Cytokines may be regarded as soluble messenger molecules of immune system. They can act as short messengers between the cells or long range messengers by circulating in the blood and affecting cells at far off sites. The latter function is comparable to that of hormones.

The term ***interleukin (IL)*** is frequently used to represent cytokines. There are more than a dozen interleukins (IL-I.....IL_{12}), produced by different cells with wide range of functions. The main function (directly or indirectly) of cytokines is to ***amplify immune responses*** and inflammatory responses.

Therapeutic uses of cytokines

It is now possible to produce cytokines *in vitro*. Some of the cytokines have potential applications in the practice of medicine. For instance, IL-2 is used in cancer immunotherapy, and in the treatment of immunodeficiency diseases. IL-2 induces the proliferation and differentiation of T-and B-cells, besides increasing the cytotoxic capacity of natural killer cells.

A group of cytokines namely interferons can combat viral infection by inhibiting their replication.

IMMUNITY IN HEALTH AND DISEASE

The prime function of imune system is to protect the host against the invading pathogens. The body tries its best to overcome various strategies of infectious agents (bacteria, viruses), and provides immunity.

Some of the important immunological aspects in human health and disease are briefly described.

AUTOIMMUNE DISEASES

In general, the immune system is self-tolerant i.e. not responsive to cells or proteins of self. Sometimes, for various reasons, the ***immune system fails to discriminate between self and non-self***. As a consequence, the cells or tissues of the body are attacked. This phenomenon is referred to as ***autoimmunity*** and the diseases are regarded as autoimmune diseases. The antibodies produced to self molecules are regarded as ***autoantibodies***. Some examples of autoimmune diseases are listed.

- Insulin-dependent diabetes (pancreatic β-cell autoreactive T-cells and antibodies).
- Rheumatoid arthritis (antibodies against proteins present in joints).
- Myasthenia gravis (acetylcholine receptor autoantibodies).
- Autoimmune hemolytic anemia (erythrocyte autoantibodies).

Mechanism of autoimmunity : It is widely accepted that autoimmunity generally occurs as a consequence of body's response against bacterial, viral or any foreign antigen. Some of the epitopes of foreign antigens are similar (homologous) to epitopes present on certain host proteins. This results in cross reaction of antigens and antibodies which may lead to autoimmune diseases.

ORGAN TRANSPLANTATION

The phenomenon of transfer of cells, tissues or organs from one site to another (in the same organism, ***autograft*** or from another organism ***allograft***) is regarded as organ transplantation. In case of humans, majority of organ transplantations are allografts (between two individuals). The term ***xenograft*** is used if tissues/organs are transferred from one species to another e.g. from pig to man (For more details on xenotransplantation Refer Chapter 41).

Organ transplantation is associated with immunological complications, and tissue rejection. This is because the host body responds to the transplanted tissue in a similar way as if it were an invading foreign organism. Major histocompatibility complex (MHC) is primarily involved in allograft rejection. This is due to the fact that MHC proteins are unique to each individual, and the immune system responds promptly to foreign MHCs.

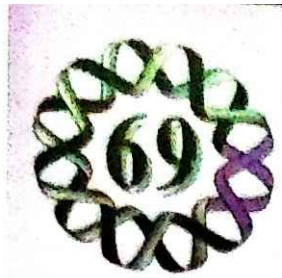
Organ transplantation between closely related family members is preferred, since their MHCs are also likely to be closely related. And major immunological complications can be averted.

CANCERS

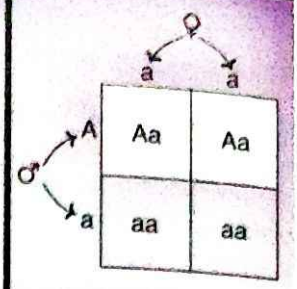
Growth of tumors is often associated with the formation novel antigens. These ***tumor antigens*** (also referred to as oncofetal antigens e.g. α-fetoprotein) are recognized as non-self by the immune systems. However, tumors have developed several mechanisms to evade immune responses.

AIDS

Acquired immunodeficiency syndrome (AIDS), caused by human immunodeficiency virus, is characterized by immunosuppression, secondary neoplasma and neurological manifestations. AIDS primarily affects the cell-mediated immune system which protects the body from intracellular parasites such as viruses, and bacteria. Most of the immuno-deficiency symptoms of AIDS are associated with a reduction in CD_4 (cluster determinant antigen 4) cells.



Genetics



Genetics is the ***study of heredity***. It is appropriately regarded as the science that explains the similarities and differences among the related organisms.

The blood theory of inheritance in humans

For many centuries, it was customary to explain inheritance in humans through blood theory. People used to believe that the children received blood from their parents, and it was the union of blood that led to the blending of characteristics. That is how the terms 'blood relations', 'blood will tell', and 'blood is thicker than water' came into existence. They are still used, despite the fact that blood is no more involved in inheritance. With the advances in genetics, the more appropriate terms should be as follows

- ***Gene relations*** in place of blood relations.
- ***Genes will tell*** instead of blood will tell.

BRIEF HISTORY AND DEVELOPMENT OF GENETICS

Genetics is relatively young, not even 150 years. The blood theory of inheritance was questioned in 1850s, based on the fact that the semen contained no blood. Thus, blood was not being transferred to the offspring. Then the big question was what was the hereditary substance.

Mendel's experiments : It was in 1866, an Austrian monk named Gregor Johann Mendel, for the first time reported the fundamental laws of inheritance. He conducted several experiments on the breeding patterns of pea plants. Mendel put forth the ***theory of transmissible factors*** which states that inheritance is controlled by ***certain factors*** passed from parents to offsprings. His results were published in 1866 in an obscure journal Proceedings of the Society of Natural Sciences.

For about 35 years, the observations made by Mendel went unnoticed, and were almost forgotten. Two European botanists (Correns and Hugo de Vries) in 1900, independently and simultaneously rediscovered the theories of Mendel. The year 1900 is important as it marks the beginning the modern era of genetics.

The origin of the word gene : In the early years of twentieth century, it was believed that the Mendel's inheritance factors are very closely related to chromosomes (literally coloured bodies) of the cells. It was in 1920s, the term ***gene*** (derived from a Greek word *gennan* meaning to produce) was introduced by Willard Johannsen. Thus, ***gene replaced the earlier terms inheritance factor or inheritance unit***.

Chemical basis of heredity : There was a controversy for quite sometime on the chemical basis of inheritance. There were two groups—the protein supporters and DNA supporters. It was in 1944, Avery and his associates presented convincing evidence that the chemical basis of heredity lies in DNA, and not in protein. Thus, ***DNA*** was finally

identified as ***the genetic material***. Its structure was elucidated in 1952 by Watson and Crick.

Importance of genes in inheritance—studies on twins

Monozygotic or identical twins contain the same genetic material — DNA or genes. Studies conducted on identical twins make starting revelations with regard to inheritance. One such study is described here.

Oskar Stohr and Jack Yufe were identical twins separated at birth. Oskar was taken to Germany where he was brought up by his grandmother as a Christian. Jack was raised by his father in Israel as a Jew. The two brothers were reunited at the age of 47. Despite the different environmental influences, their behavioural patterns and personalities were remarkably similar

- Both men had moustaches, wore two pocket shirts, and wire-rimmed glasses.
- Both loved spicy foods and tended to fall asleep in front of television.
- Both flushed the toilet before using.
- Both read maganizes from back to front.
- Both stored rubber bands on their wrists.
- Both liked to sneeze in a room of strangers.

Besides Oskar and Jack, many other studies conducted on identical twins point out the importance of genes on the inherited characters related to personality and mannerisms.

BASIC PRINCIPLES OF HEREDITY IN HUMANS

The understanding of how genetic characteristics are passed on from one generation to the next is based on the principles developed by Mendel.

As we know now, the human genome is organized into a diploid (2n) set of 46 chromosomes. They exist as ***22 pairs of autosomes*** and ***one pair of sex chromosomes (XX/XY)***. During the course of meiosis, the chromosome number becomes haploid (n). Thus, haploid male and female gametes — sperm and oocyte respectively, are formed. On fertilization of the oocyte by the sperm, the diploid status is restored. This becomes possible as the zygote receives one member of each chromosome pair from the father, and the other from the mother. As regards the sex chromosomes, the males have X and Y, while the females have XX. The sex of the child is determined by the father.

Monogenic and polygenic traits

The genetic traits or characters are controlled by single genes or multiple genes. The changes in genes are associated with genetic diseases.

Monogenic disorders : These are the single gene disease traits due to alterations in the corresponding gene e.g. Sickle-cell anemia, phenylketonuria. Inheritance of monogenic disorders usually follows the Mendelian pattern of inheritance.

Polygenic disorders : The genetic traits conferred by more than on gene (i.e multiple genes), and the disorders associated with them are very important e.g. height, weight, skin colours, academic performance, blood pressure, aggressiveness, length of life.

PATTERNS OF INHERITANCE

The heredity is transmitted from parent to offspring as individual characters controlled by genes. The genes are linearly distributed on chromosomes at fixed positions called ***loci***.

A gene may have different forms referred to as ***alleles***. Usually one allele is transferred from the father, and the other from the mother. The allele is regarded as ***dominant*** if the trait is exhibited due to its presence. On the other hand, the allele is said to be ***recessive*** if its effect is masked by a dominant allele. The individuals are said to be ***homozygous*** if both the alleles are the same. When the alleles are different they are said to be ***heterozygous***.

The pattern of inheritance of monogenic traits may occur in the following ways (***Fig. 69.1***).

1. Autosomal dominant
2. Autosomal recessive
3. Sex-linked.

1. **Autosomal dominant inheritance :** A normal allele may be designated as **a** while an autosomal dominant disease allele as ***A*** (***Fig. 69.1A***). The male with Aa genotype is an affected one while the female with aa is normal. Half of the genes from the affected male will carry the disease allele. On mating, the male and female gametes are mixed in different combinations. The result is that ***half of the children*** will be heterozygous (Aa) and ***have the disease***. Example of autosomal dominant inherited diseases are familial hypercholesterolemia, β-thalassemia, breast cancer genes.

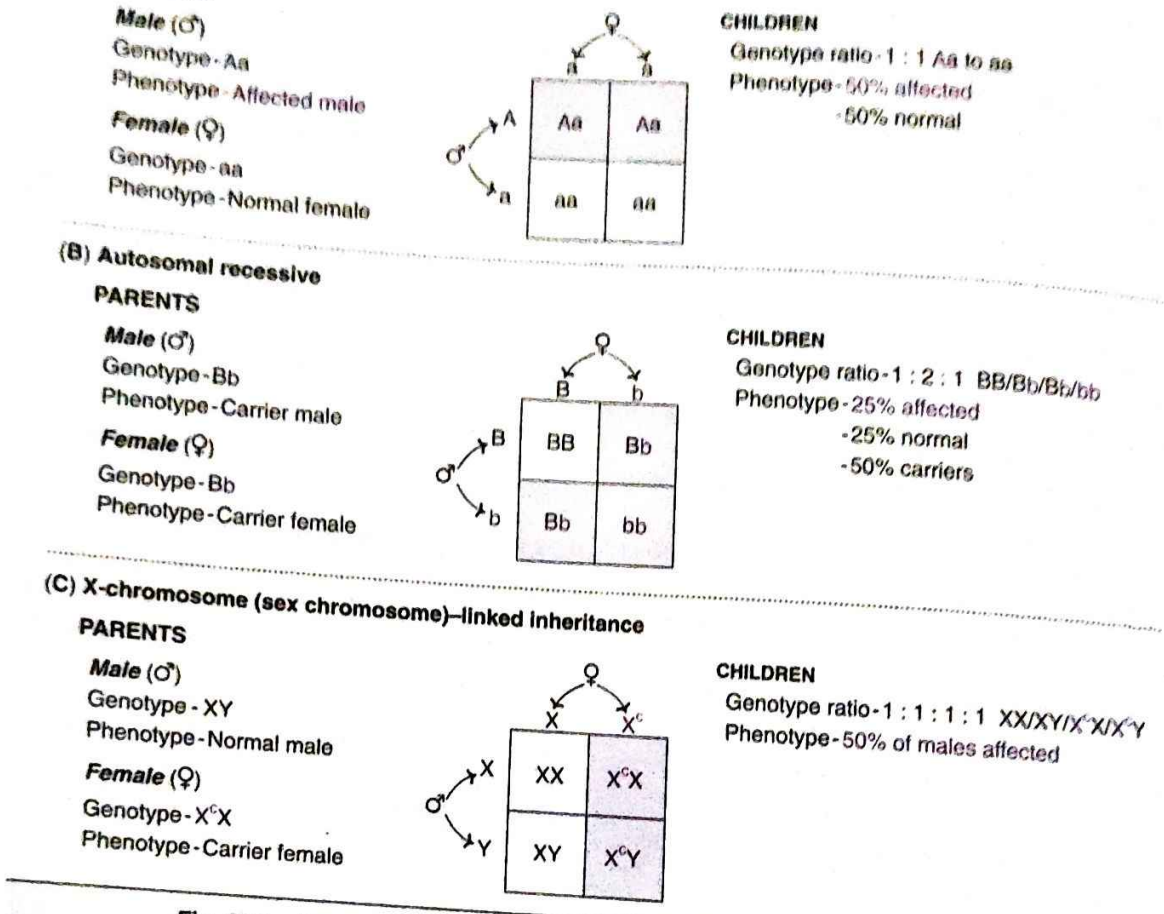


***Fig. 69.1** : Patterns of inheritance-autosomal dominant, autosomal recessive and X-linked (**Note** : Genotype refers to the description of genetic composition, while phenotype represents the observable character displayed by an organism).*

2. **Autosomal recessive inheritance :** In this case, the normal allele is designated as ***B*** while the disease-causing one is ***a*** (***Fig. 69.1B***). The gametes of carrier male and carrier female (both with genotype Bb) get mixed. For these heterozygous carrier parents, there is one fourth chance of having an affected child. Cystic fibrosis, sickle-cell anemia and phenylketonuria are some good examples of autosomal recessive disorders.

3. **Sex (X)-linked inheritance :** In the ***Fig. 69.1C***, sex-linked pattern of inheritance is depicted. A normal male (XY) and a carrier female (X^cY) will produce children wherein, half of the male children are affected while no female children is affected. This is due to the fact that the male children possess only one X chromosome, and there is no dominant allele to mark its effects (as is the case with females). Colour blindness and hemophilia are good examples of X-linked diseases.

A selected list of genetic disorders (monogenic traits) due to autosomal and sex-linked inheritance in humans is given in ***Table 69.1***.

GENETIC DISEASES IN HUMANS

The pattern of inheritance and monogenic traits along with some of the associated disorders are described above (***Table 69.1***). Besides ***gene mutations***, ***chromosomal abnormalities*** (aberrations) also result in genetic diseases.

Table 69.1 Selected examples of genetic disorders (monogenic traits) in humans

Inherited pattern/disease	*Estimated incidence*	*Salient features*
Autosomal dominant		
Familial hypercholesterolemia	1 : 500	High risk for heart diseases
Huntington disease	1 : 5000	Nervous disorders, dementia
Familial retinoblastoma	1 : 12000	Tumors of retina
Breast cancer genes *(BRAC 1 and 2)*	1 : 800	High risk for breast and ovarian cancers
β-Thalassemia	1 : 2500 (in people of Mediterranean descent)	A blood disorder; the blood appears to be blue instead of red
Autosomal recessive		
Sickle-cell anemia	1 : 100 (in Africans)	Severe life threatening anemia; confers resistance to malaria
Cystic fibrosis	1 : 2500 (in Caucasians)	Defective ion transport; severe lung infections and early death (before they reach 30 years)
Phenylketonuria	1 : 2000	Mental retardation due to brain damage
α_1-Antitrypsin deficiency	1 : 5000	Damage to lungs and liver
Tay-Sachs disease	1 : 3000 (in Ashkenazi Jews)	Nervous disorder; blindness and paralysis
Severe combined immunodeficiency disease (SCID)	Rare (only 100 cases reported worldwide)	Highly defective immune system; early death
Sex-linked		
Colour blindness	1 : 50 males	Unable to distinguish colours
Hemophilia (A/B)	1 : 10,000 males	Defective blood clotting
Duchenne muscular dystrophy	1 : 7000 males	Muscle wastage
Mitochondrial		
Leber hereditary optic neuropathy	Not known	Damage to optic nerves, may lead to blindness

Aneuploidy : The presence of ***abnormal number of chromosomes*** within the cells is referred to as aneuploidy. The most common aneuploid condition is ***trisomy*** in which three copies of a particular chromosome are present in a cell instead of the normal two e.g. trisomy-21 causing ***Down's syndrome***; trisomy-18 that results in ***Edward's syndrome***. These are the examples of autosomal aneuploidy.

In case of sex-linked aneuploidy, the sex chromosomes occur as three copies. e.g. phenotypically male causing Klinefelter's syndrome has XXY; trisomy-X is phenotically a female with XXX.

Selected examples of chromosomal disorders along the with the syndromes and their characteristic features are given in ***Table 69.2***.

EUGENICS

Eugenics is a ***science of improving human race based on genetics***. Improving the traits of plants and animals through breeding programmes has been in practice for centuries.

Eugenics is a highly controversial subject due to social, ethical, and political reasons. The proponents of eugenics argue that people with

Table 69.2 Selected examples of chromosomal disorders in humans

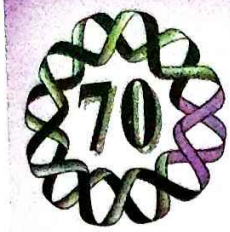
Condition	*Chromosome*	*Syndrome*	*Salient features*
Normal			
Females	2n (46, XX)	—	—
Males	2n (46, XY)	—	—
Autosomal aneuploidy	**2n + specific chromosome**		
Trisomy-21	47, XY, +21	Down's syndrome	IQ<50; Congenital heart disease, low life expectancy
Trisomy-18	47, XY, +21	Edward's syndrome	Malformation of heart, kidney and other organs
Trisomy-13	47, XY, +13	Patau's syndrome	Small head, polydactyly, congenital heart diseases
Sex-linked aneuploidy			
Phenotypically male	47, XXY	Klinefelter's syndrome	Small testes, infertility
	47, XYY	Jacob's syndrome	Usually asymptomatic, often tall
Phenotically female	47, XXX	Triplo-X (trisomy-X)	Usually asymptomatic, about 20% mentaly handicapped
	47, XO (missing Y)	Turner's syndrome	Short stature.

desirable and good traits (good blood) should reproduce while those with undersirable characters (bad blood) should not. The advocates of eugenics, however, do not force any policy, but they try to convince the people to perform their duty voluntarily. The object of eugenics is to limit the production of people who are unfit to live in the society.

Eugenics in Nazi Germany

Germany developed its own eugenic programme during 1930s. A law on eugenic sterilization was passed in 1933. In a span of three years, compulsory sterilization was done on about 250,000 people, who allegedly suffered from hereditary disabilities, feeble mindedness, epilepsy, schizophrenia, blindness, physical deformaties, and drug or alcohol addition.

The German Government committed many atrocities in the name of racial purity. Other countries, however do not support this kind of eugenics.



70 Bioinformatics



Bioinformatics is ***the combination (or marriage!) of biology and information technology***. Basically, bioinformatics is a recently developed science using information to understand biological phenomenon. It broadly involves the computational tools and methods used to manage, analyse and manipulate volumes and volumes of biological data.

Bioinformatics may also be regarded as a part of the ***computational biology***. The latter is concerned with the application of quantitative analytical techniques in modeling and solving problems in the biological systems. Bioinformatics is an interdisciplinary approach requiring advanced knowledge of computer science, mathematics and statistical methods for the understanding of biological phenomena at the molecular level.

History and relevance of bioinformatics

The term bioinformatics was first introduced in 1990s. Originally, it dealt with the management and analysis of the data pertaining to ***DNA, RNA and protein sequences***. As the biological data is being produced at an unprecedented rate, its management and interpretation invariably requires bioinformatics. Thus, bioinformatics now includes many other types of biological data. Some of the most important ones are listed below

- Gene expression profiles
- Protein structure
- Protein interactions
- Microarrays (DNA chips)
- Functional analysis of biomolecules
- Drug designing.

Bioinformatics is largely (not exclusively) a computer-based discipline. Computers are in fact very essential to handle large volumes of biological data, their storage and retrieval.

We have to accept the fact that there is no computer on earth (however advanced) which can store information, and perform the functions like a living cell. Thus a highly complex information technology lies right within the cells of an organism. This primarily includes the organism's genes and their dictates for the organisms biological processes and behaviour.

BROAD COVERAGE OF BIOINFORMATICS

Bioinformatics covers many specialized and advanced areas of biology.

Functional genomics : Identification of genes and their respective functions.

Structural genomics : Predictions related to functions of proteins.

Comparative genomics : For understanding the genomes of different species of organisms.

DNA microarrays : These are designed to measure the levels of gene expression in different tissues, various stages of development and in different diseases.

Medical informatics : This involves the management of biomedical data with special referece to biomolecules, *in vitro* assays and clinical trials.

COMPONENTS OF BIOINFORMATICS

Bioinformatics comprises three components

1. **Creation of databases** : This involves the organizing, storage and management the biological data sets. The databases are accessible to researchers to know the existing information and submit new entries. e.g. protein sequence data bank for molecular structure. Databases will be of no use until analysed.

2. **Development of algorithms and statistics** : This involves the development of tools and resources to determine the relationship among the members of large data sets e.g. comparison of protein sequence data with the already existing protein sequences.

3. **Analysis of data and interpretation** : The appropriate use of components 1 and 2 (given above) to analyse the data and interpret the results in a biologically meaningful manner. This includes DNA, RNA and protein sequences, protein structure, gene expression profiles and biochemical pathways.

BIOINFORMATICS AND THE INTERNET

The internet is an ***inter***national computer ***net***work. A computer network involves a group of computers that can communicate (usually over a telephone system) and exchange data between users.

It is the ***internet protocol*** (IP) that determines how the packets of information are addressed and routed over the network. To access the internet, a computer must have the correct hardware (modem/ network card), appropriate software and permission for access to network. For this purpose, one has to subscribe to an ***internet service provider*** (ISP).

World wide web (www) : www involves the exchange of information over the internet using a programme called ***browser***. The most widely used browsers are Internet explorer and Netscape navigator.

www works on the basis of ***Uniform resource locator*** (URL) which is a document with a unique address. URLs takes the format ***http.//*** (***h***yper***t***ext ***t***ransfer ***p***rotocol) that can identify the protocol for communication over www.

BIOLOGICAL DATABASES

The collection of the ***biological data*** on a computer which can be manipulated to appear in ***varying arrangements and subsets*** is regarded as a database. The biological information can be stored in different databases. Each database has its own website with unique navigation tools.

The biological databases are, in general, publicly accessible. Selected examples of biological databases are briefly described (***Table 70.1***).

Nucleotide sequence databases

The nucleotide sequence data submitted by the scientists and genome sequencing groups is at the databases namely GenBank, EMBL (European Molecular Biology Laboratory) and DDBJ (DNA Data Bank of Japan). There is a good coordination between these three databases as they are synchronized on daily basis.

Besides the primary nucleotide databases (referred above), there are some other databases also to provide information on genes, genomes and ongoing research projects.

Protein sequence databases

Protein sequence databases are usually prepared from the existing literature and/or in consultation with the experts. In fact, these databases represent the translated DNA databases.

Molecular structure of databases

The three dimensional (3-D) structures of macromolecules are determined by X-ray crystallography and nuclear magnetic resonance (NMR). PDB and SCOP are the primary databases of 3-D structures of biological molecules.

Other databases

KEGG database is an important one that provides information on the current knowledge of molecular biology and cell biology with special reference to information on metabolic pathways, interacting molecules and genes.

Table 70.1 Selected examples of biological databases in bioinformatics

Database	*Salient features*
Primary nucleotide sequence databases	
GenBank (www.ncbi.nih.gov/GeneBank/)	Provides nucleotide sequence databases maintained by the National Center for Biotechnology Information (NCBI), USA.
EMBL (www.ebi.ac.uk/embl/)	European Molecular Biology Laboratory (EMBL) maintains nucleotide sequence databases under the aegis of European Bioinformatics Institue (EBI), UK.
Other nucleotide sequence databases	
UniGene (www.ncbi.nih.gov/UniGene/)	The nucleotide sequences of GenBank in the form of clusters, representing genes are available.
Genome Biology (www.ncbi.nlm.nih.gov/Genomes/)	The information about the completed genomes is available.
EBI Genomes (www.ebi.uk./genomes/)	Provides data and statistics for the completed genomes, besides the information on the ongoing projects.
Protein sequence database	
SWISS-PROT (www.expasy.ch/sport)	Provides the description of the structure of a protein, its domains structure, post-translational modifications, variants etc. It has high level of integration with other databases and minimal level of redundancy.
PIR (pir.georgetown.edu)	Protein information resource (PIR) is a database provided by the National Biomedical Research Foundation (NBRF) in USA.
Protein sequence motif databases	
PROSITE (www.expasy.ch/prosite/)	Provides information on protein families and domains. It also has patterns and profiles for sequences and biological functions.
Pfam (www.cgr.ki.se/Pfam/)	A database of protein families defined in the form of domains. It has multiple alignment of a set of sequences.
Macromolecular databases	
PDB (www.rcsb.org/pdb)	This is the primary database for 3-dimensional (3-D) structures of biological macromolecules (determined by X-ray and NMR studies).
SCOP (www.mrc.lmb.cam.ac.uk/scop/)	Provides information on the structural classification of proteins (SCOP). The proteins are classified on the basis of 3-D structures.
Other databases	
KEGG (www.genome.ad.jp/kegg/)	The Kyoto Encyclopedia of Genes and Genomes (KEGG) is a database with latest computerised information on biomolecules and cell biology. KEGG provides details on information pathways, interacting molecules and the connecting links with genes.

APPLICATIONS OF BIOINFORMATICS

The advent of bioinformatics has revolutionized the advancements in biological science. And biotechnology is largely benefited by bioinformatics. The best example is the sequencing of human genome in a record time which would not have been possible without bioinformatics. A selected list of applications of bioinformatics is given below.

- Sequence mapping of biomolecules (DNA, RNA, proteins).
- Identification of nucleotide sequences of functional genes.
- Finding of sites that can be cut by restriction enzymes.
- Designing of primer sequence for polymerase chain reaction.
- Prediction of functional gene products.
- To trace the evolutionary trees of genes.
- For the prediction of 3-dimensional structure of proteins.
- Molecular modelling of biomolecules.
- Designing of drugs for medical treatment.
- Handling of vast biological data which otherwise is not possible.
- Development of models for the functioning various cells, tissues and organs.

The above list of applications however, may be treated as incomplete, since at present there is no field in biological sciences that does not involve bioinformatics.